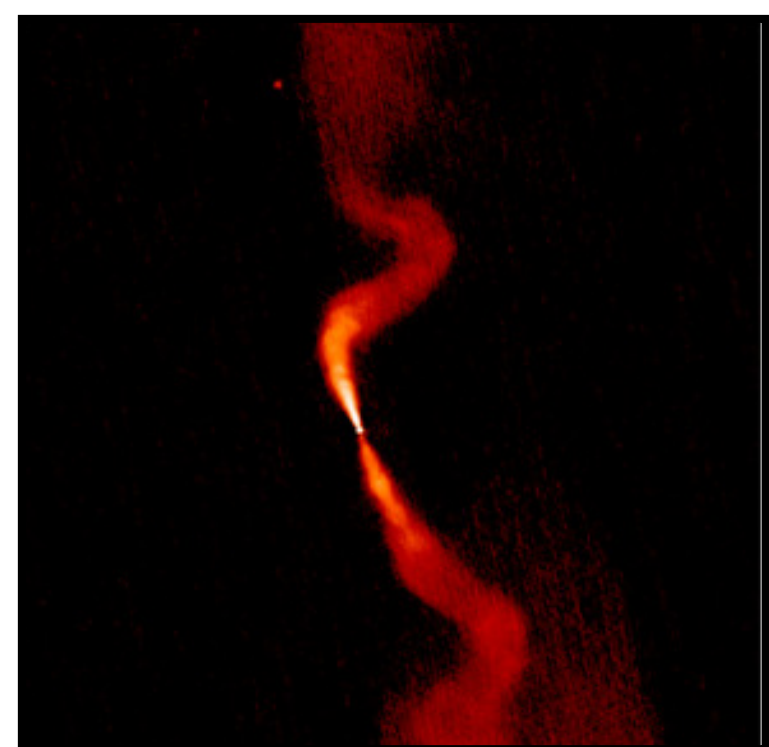


Equivariant Neural Networks

Anna Scaife - Jodrell Bank Centre for Astrophysics

Talk overview

- Invariance and astronomy
- Convolutions as translation equivariant feature extractors
- What about other types of equivariance?
- Group equivariant / G-steerable convolutions
- Why it matters in astrophysics



flatten

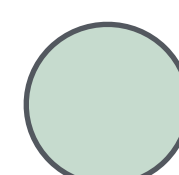
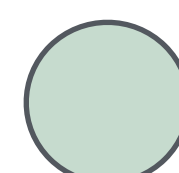
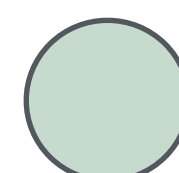
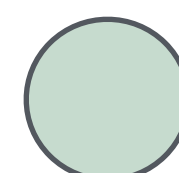
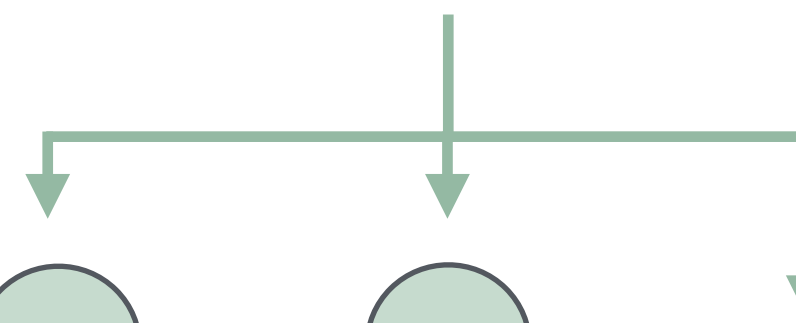
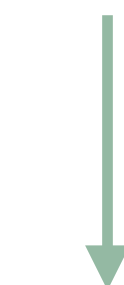
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ \vdots \\ \vdots \\ x_n \end{bmatrix}$$

features

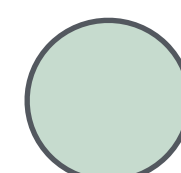
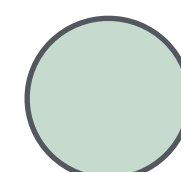
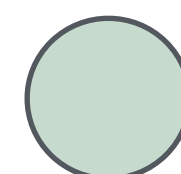
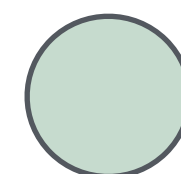
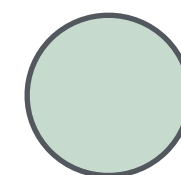
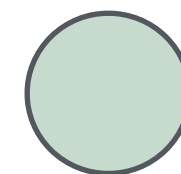
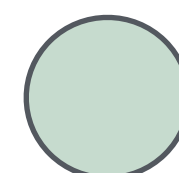
input layer

hidden layers

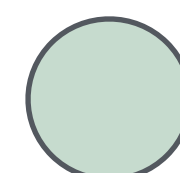
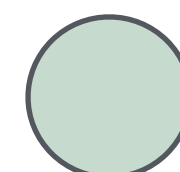
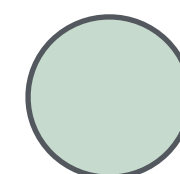
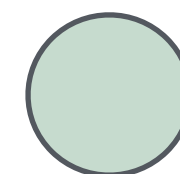
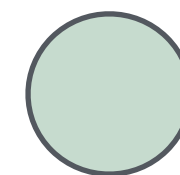
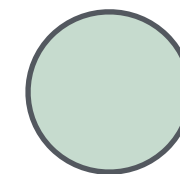
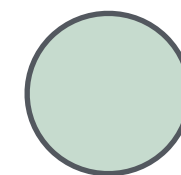
output layer



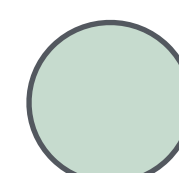
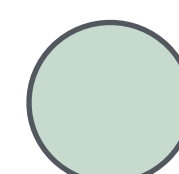
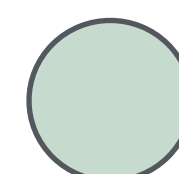
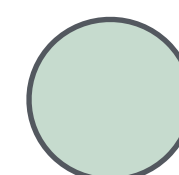
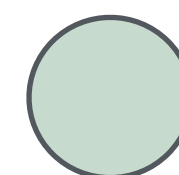
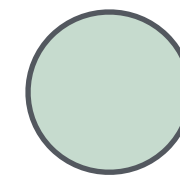
$\vdots$



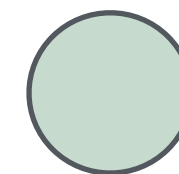
$\vdots$



$\vdots$



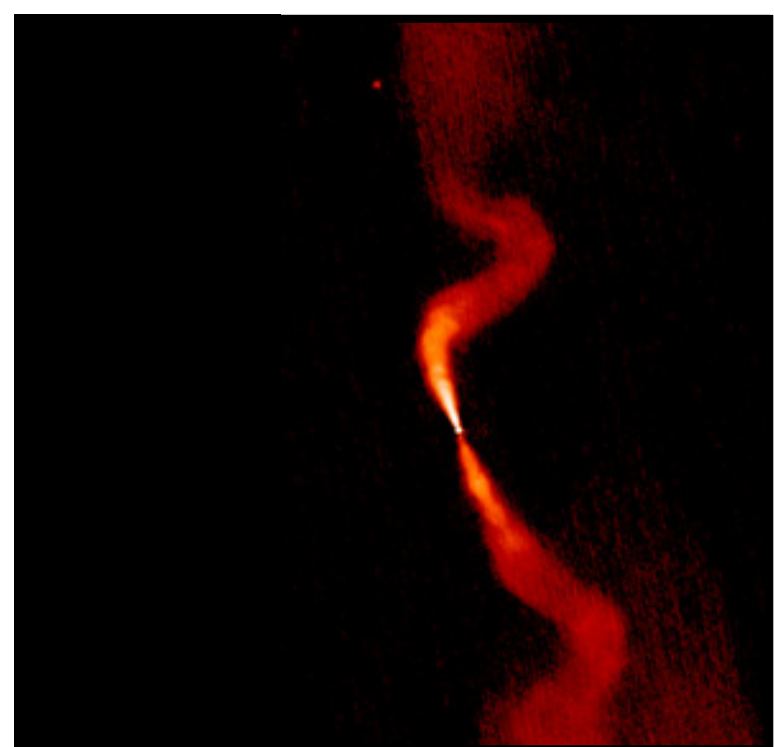
$\vdots$



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

outputs

classifier



flatten

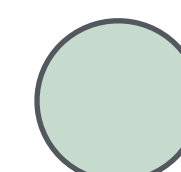
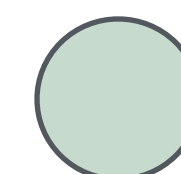
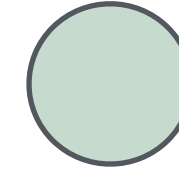
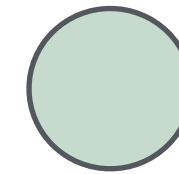
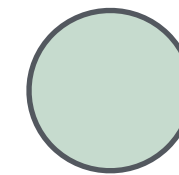
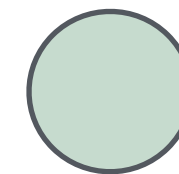
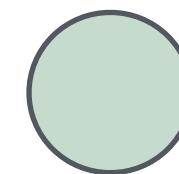
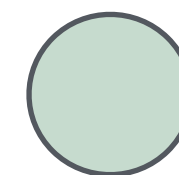
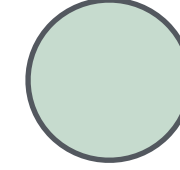
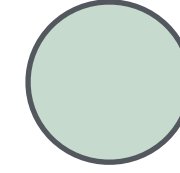
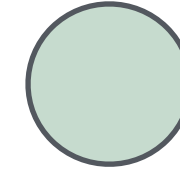
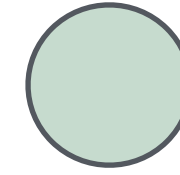
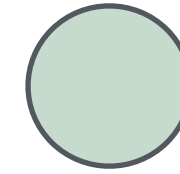
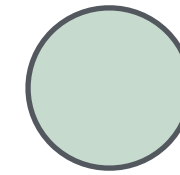
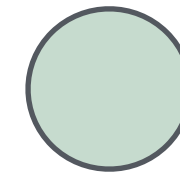
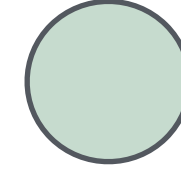
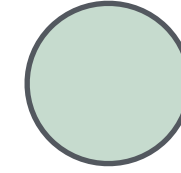
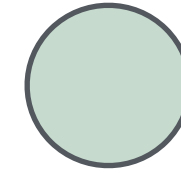
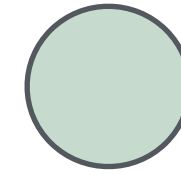
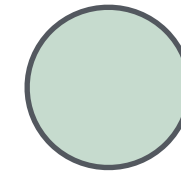
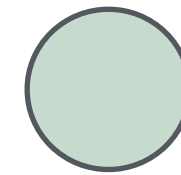
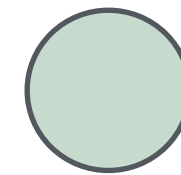
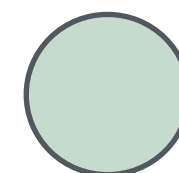
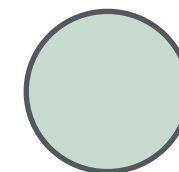
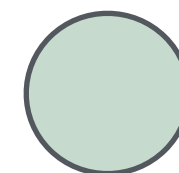
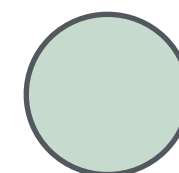
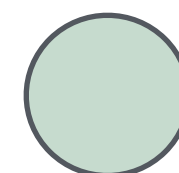
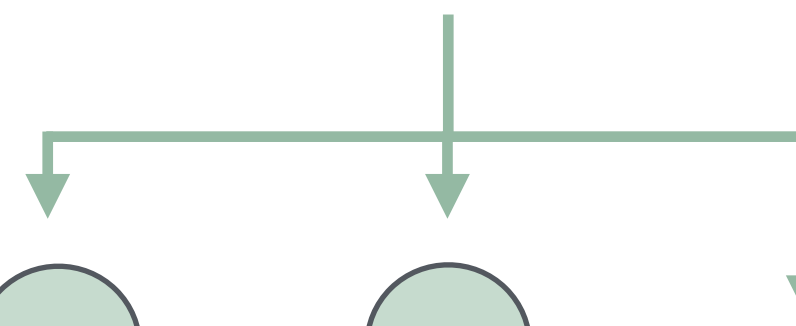
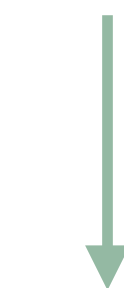
$$\begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \\ x'_4 \\ \vdots \\ \vdots \\ x'_n \end{bmatrix}$$

features

input layer

hidden layers

output layer

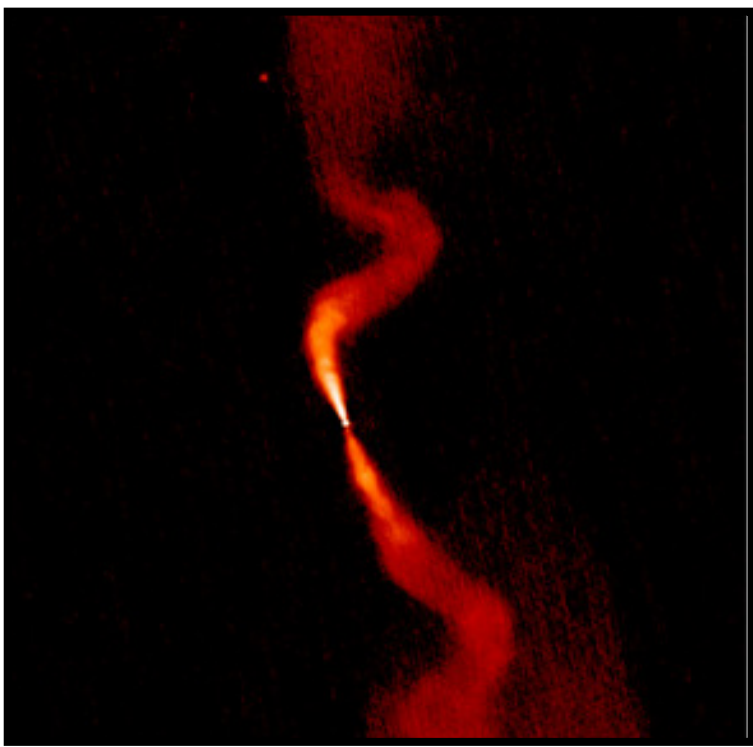
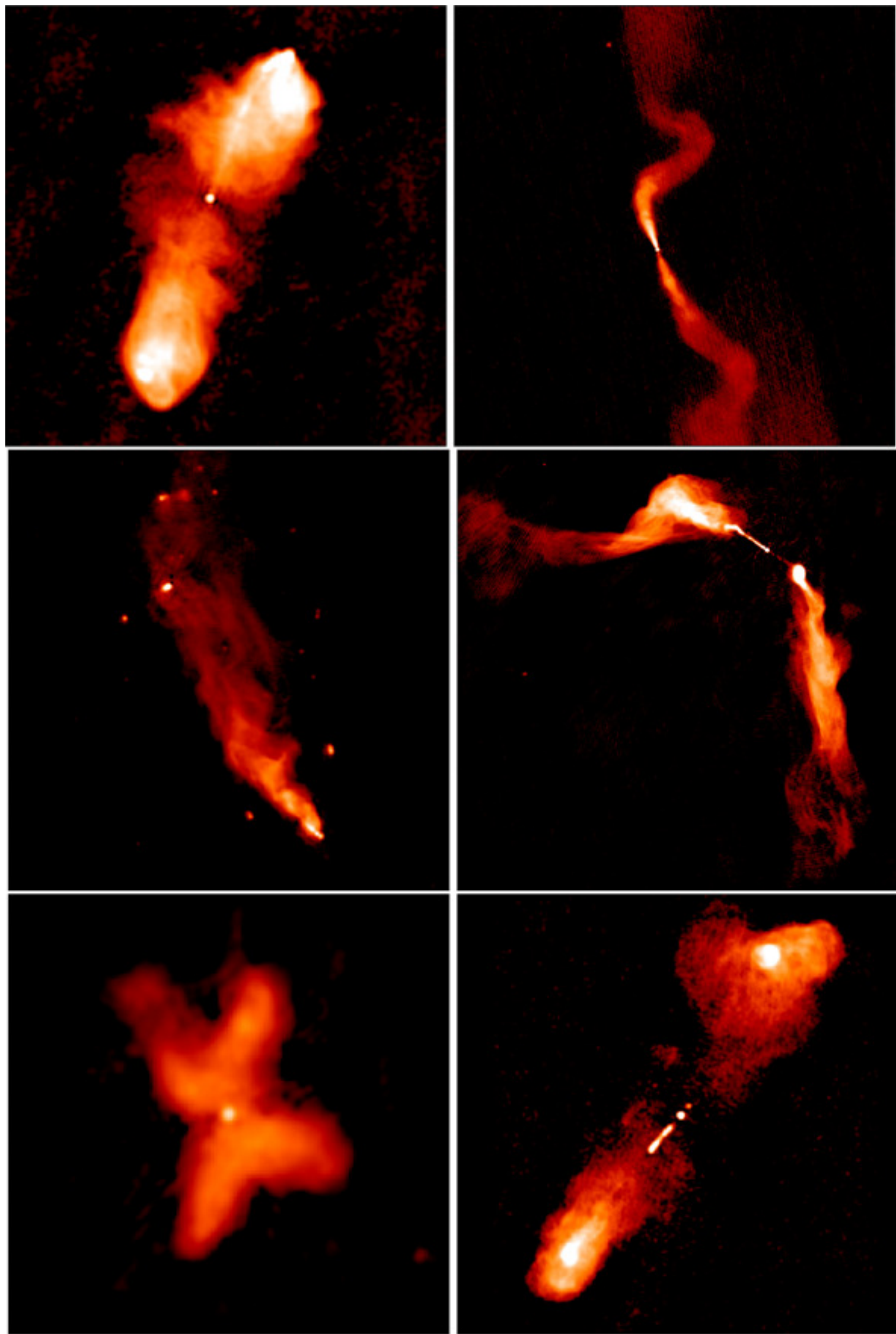


$$\begin{bmatrix} y'_1 \\ y'_2 \end{bmatrix}$$

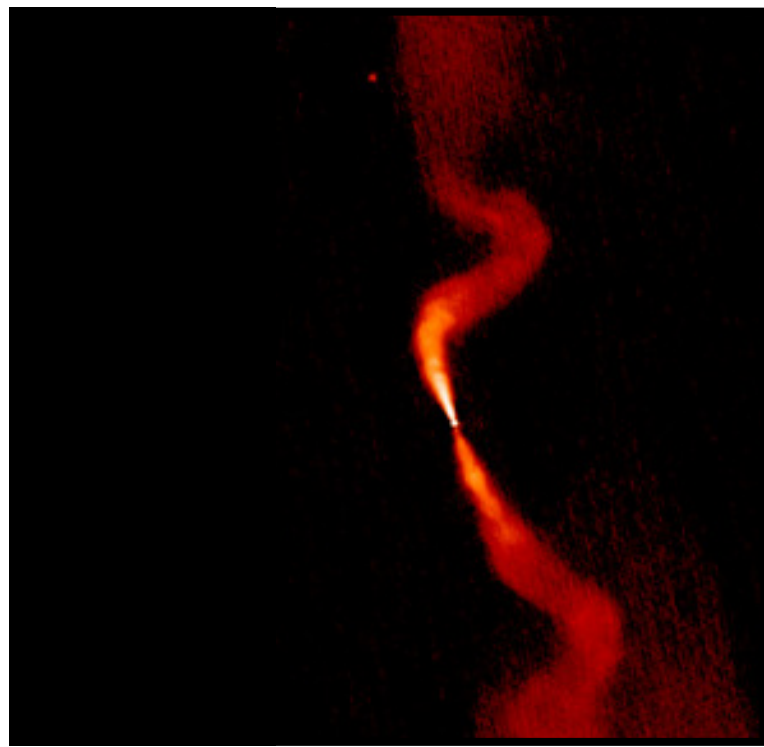
outputs

classifier

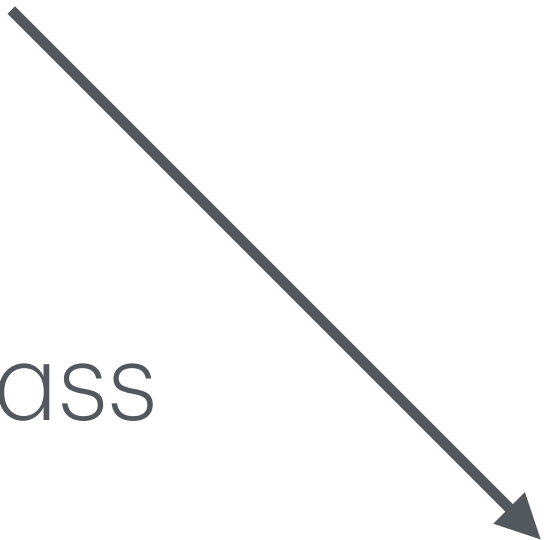
invariance



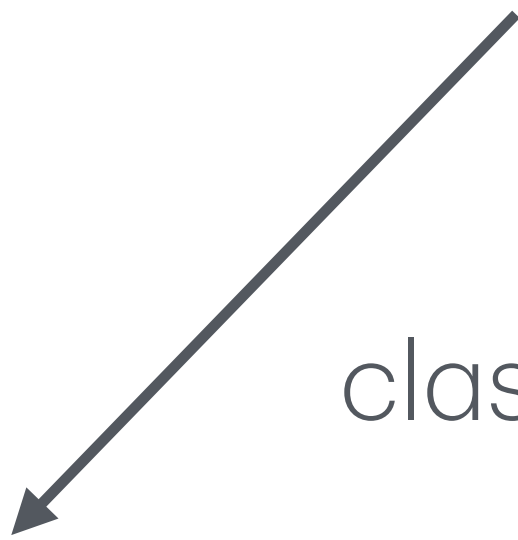
translate



class

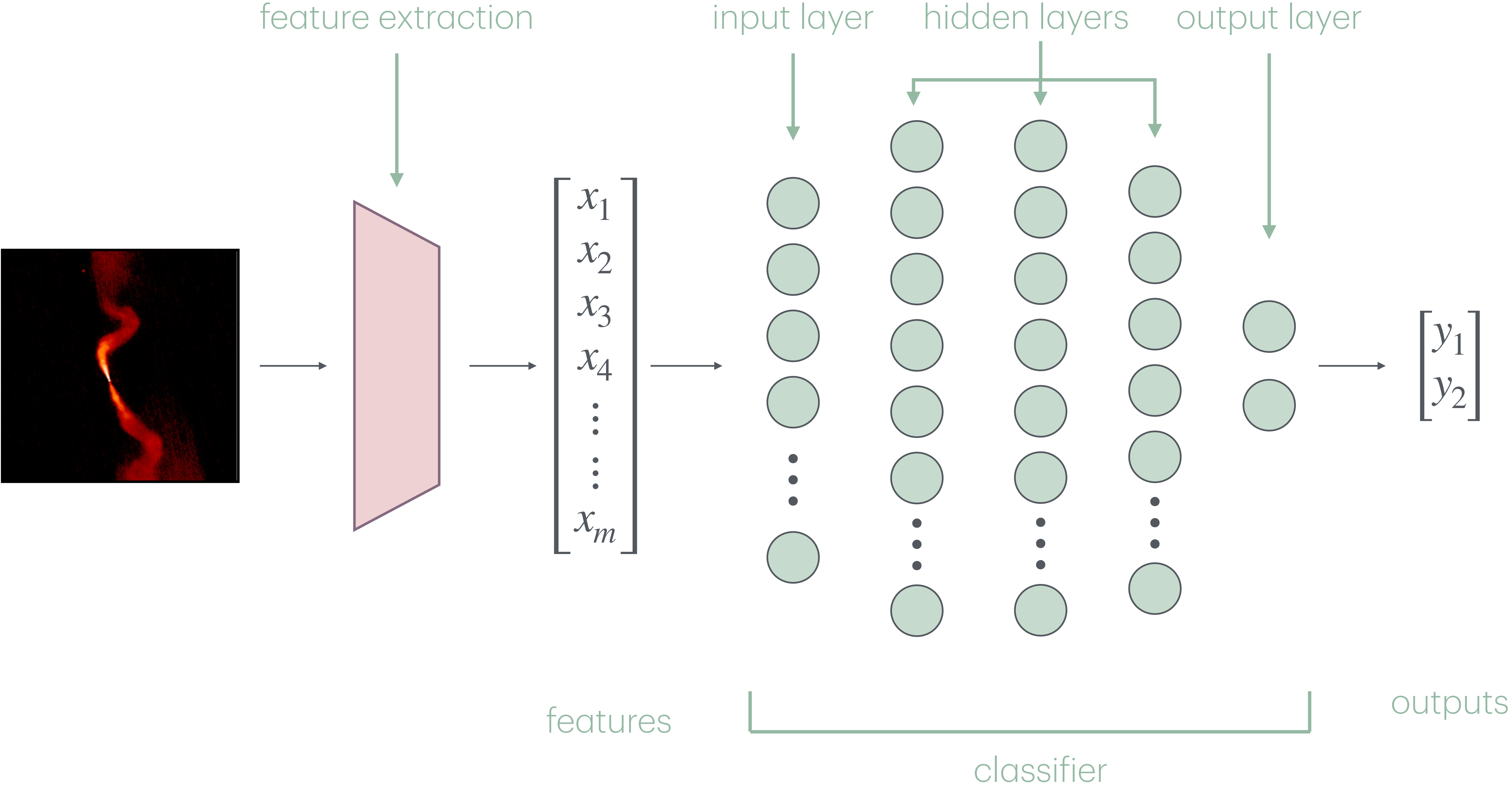


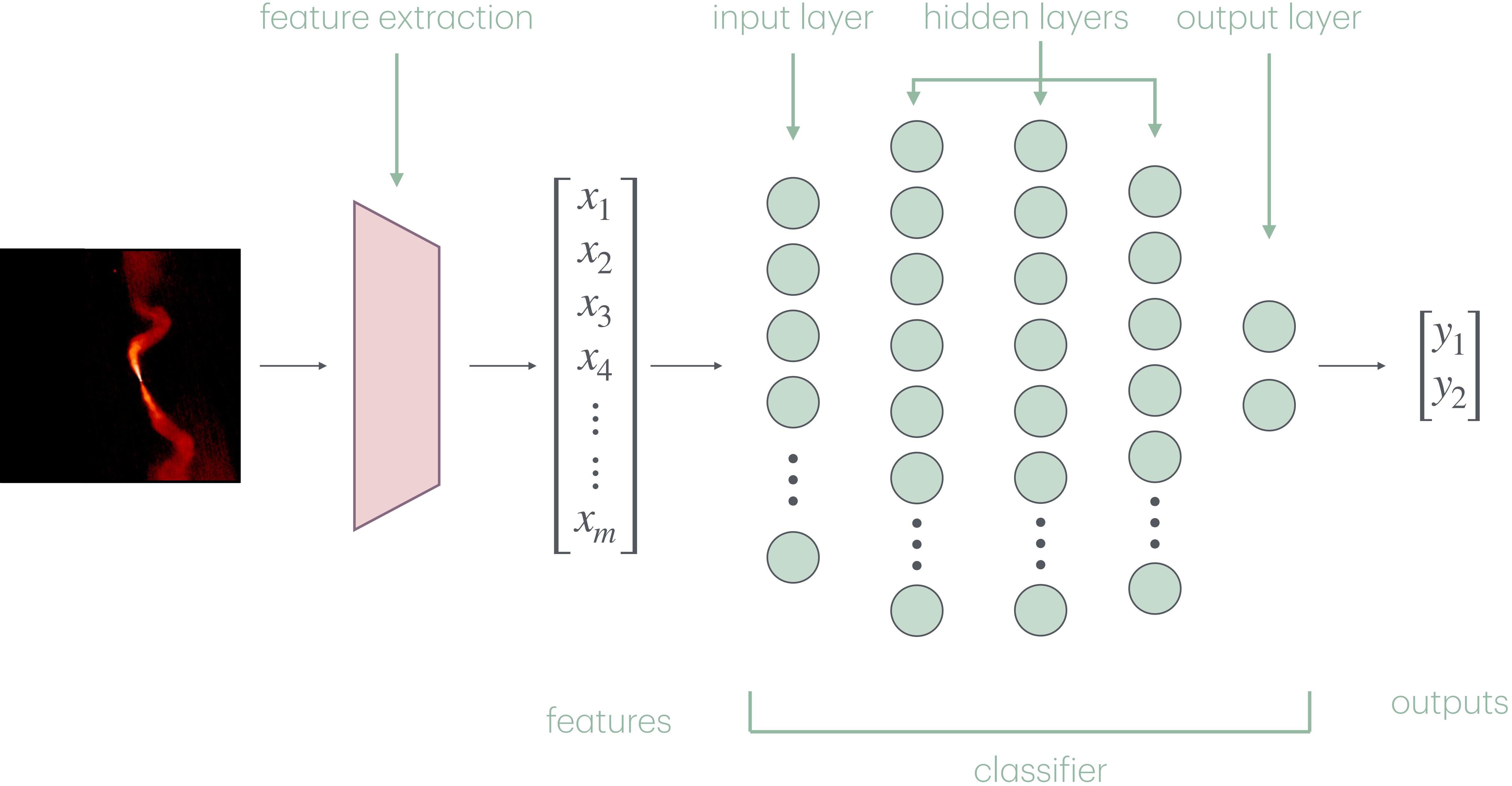
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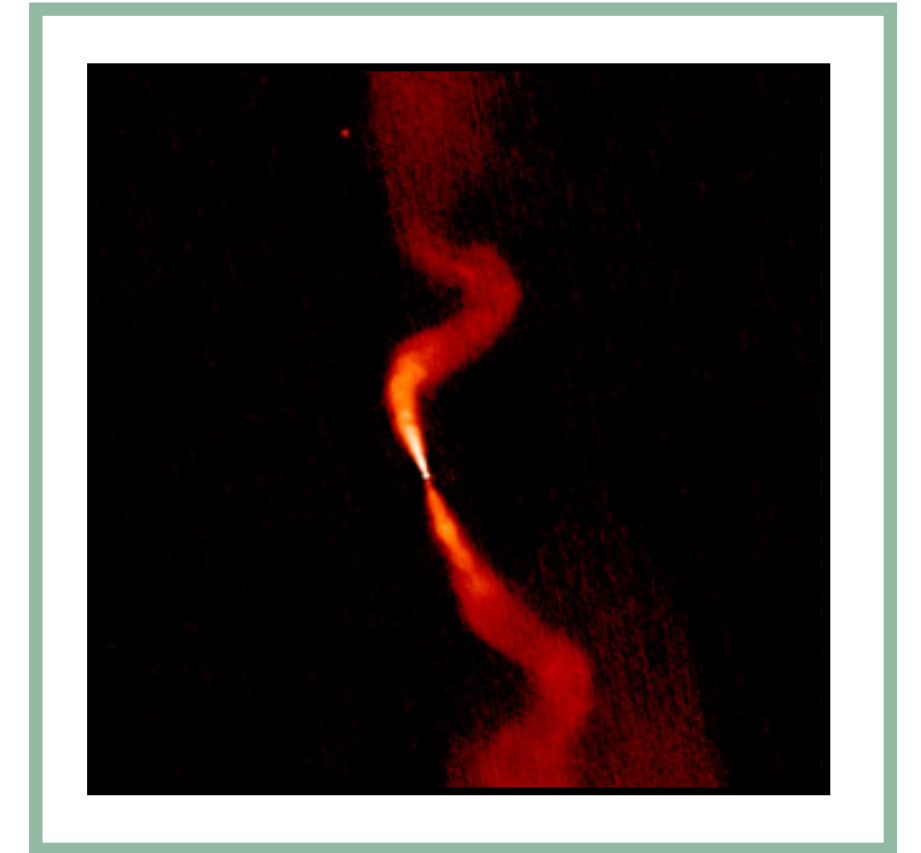
FR I







$d = 2, c = 1$



At each pixel $x \in \mathbb{R}^d$ there is a set of features, $F(x) \in \mathbb{R}^c$.

Say we have a function:

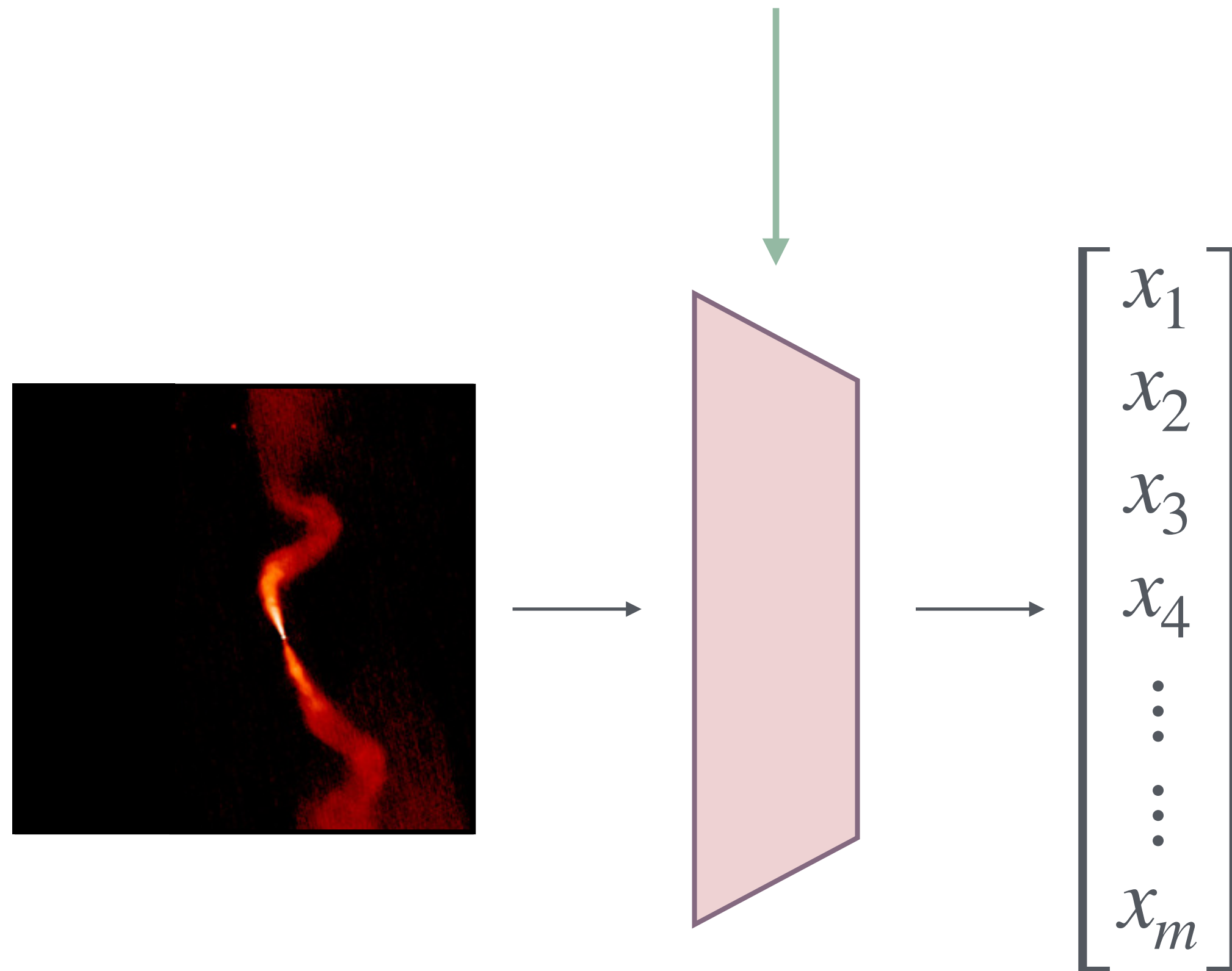
$$\mathcal{L}(F) : \mathbb{R}^d \rightarrow \mathbb{R}^{c_{\text{out}}}$$

i.e. at each pixel it takes a set of input features with dimension c_{in} and produces a set of output features with dimension c_{out} .

The function is **invariant** to translations $t \in (\mathbb{R}^d, +)$ if

$$\mathcal{L}(t \triangleright F) = \mathcal{L}(F).$$

feature extraction



Some examples:

- **Minkowski Functionals**

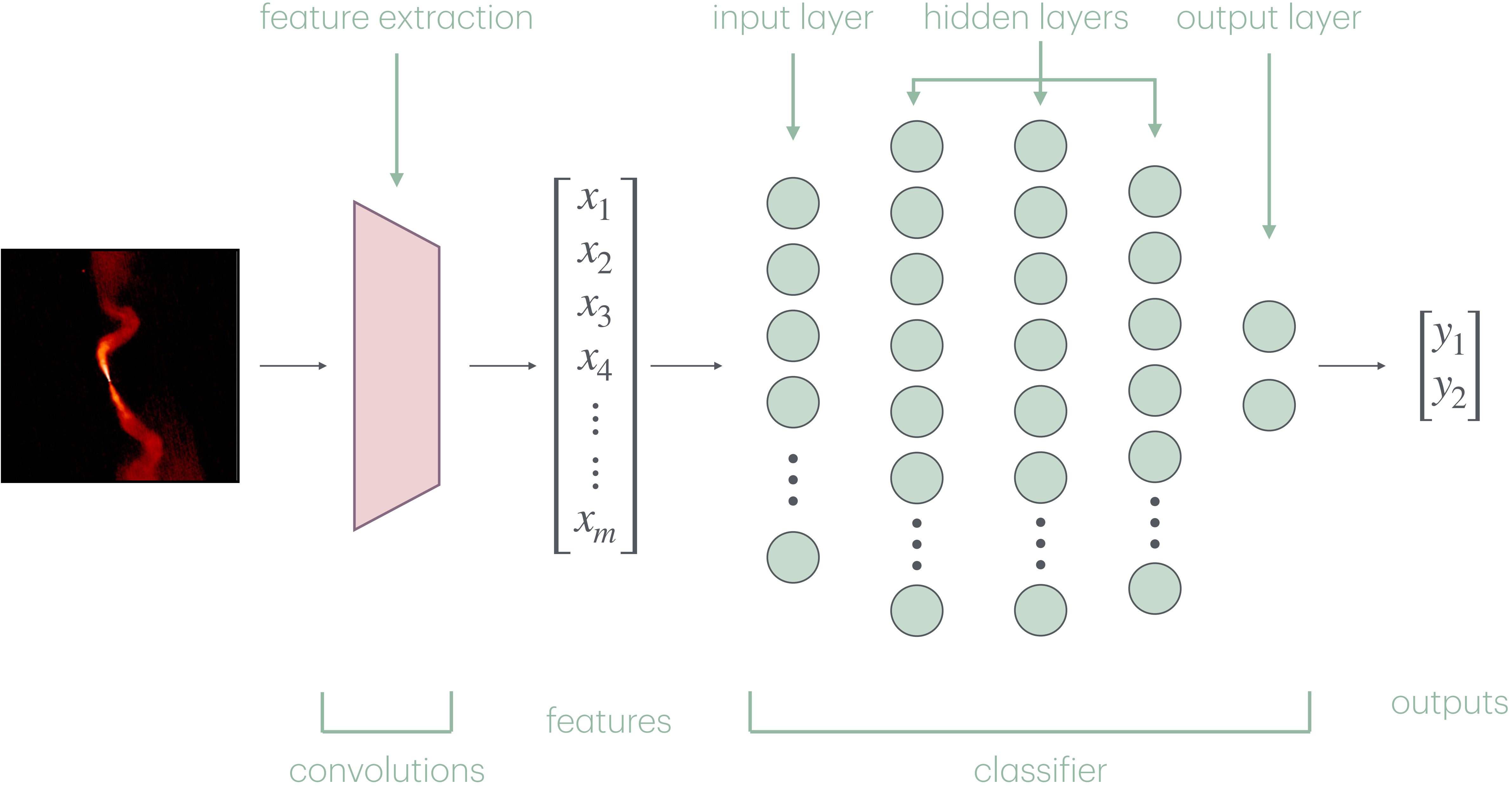
- Munshi+ 2012 (weak lensing)
- Grewal+ 2022 (galaxy clustering)
- Schmalzing & Gorski+ 1998 (CMB)

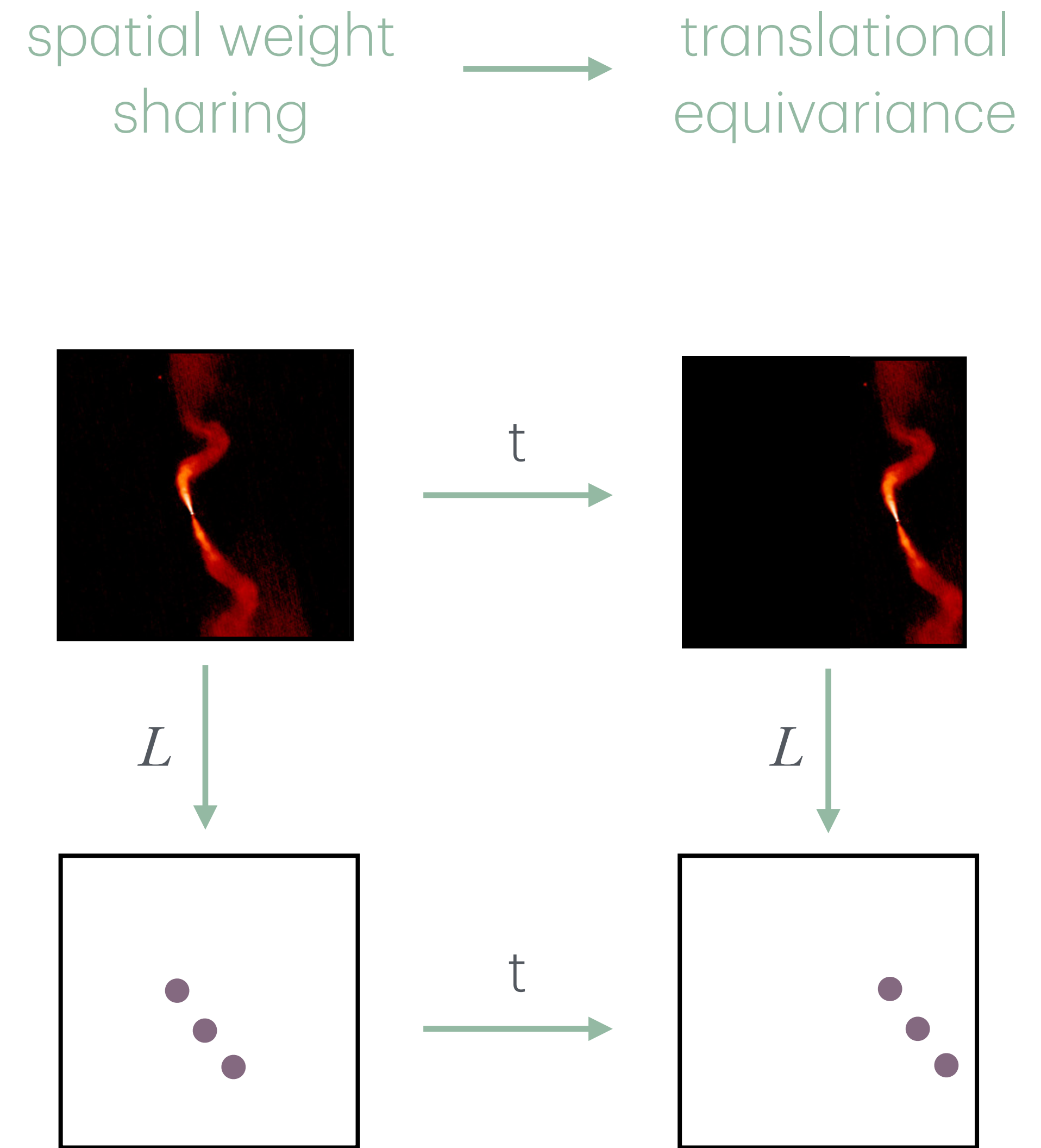
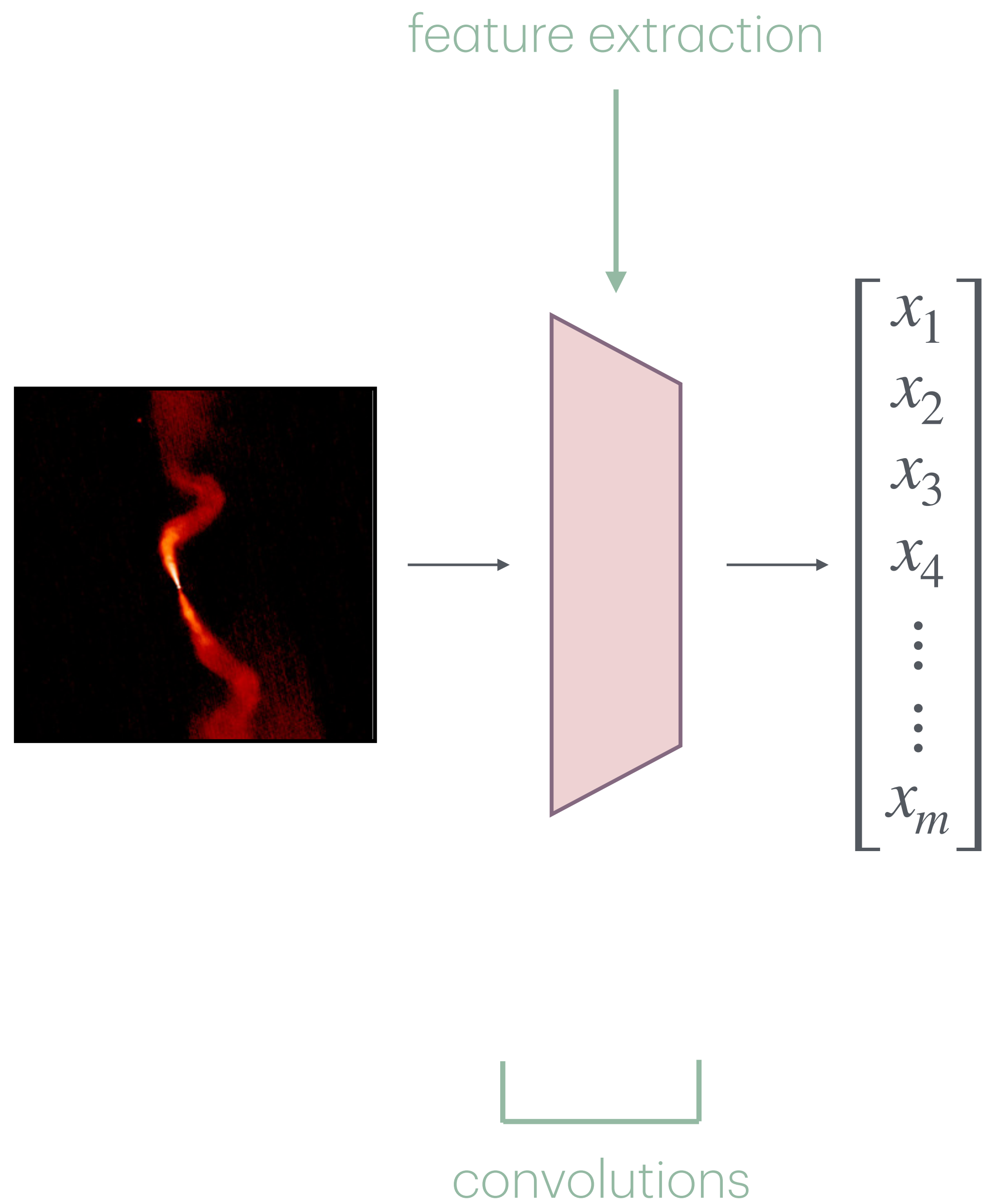
- **Haralick Features**

- Ntwaetsile & Geach 2021 (radio galaxy classification)
- ++medical literature

- **Elliptical Fourier Descriptors**

- Dhingra+ 2019 (lunar surface mapping)
- Dutta+ 2013 (Hubble sequence classification)





$d = 2, c = 1$

At each pixel $x \in \mathbb{R}^d$ there is a set of features, $F(x) \in \mathbb{R}^c$.

Say we have a function:

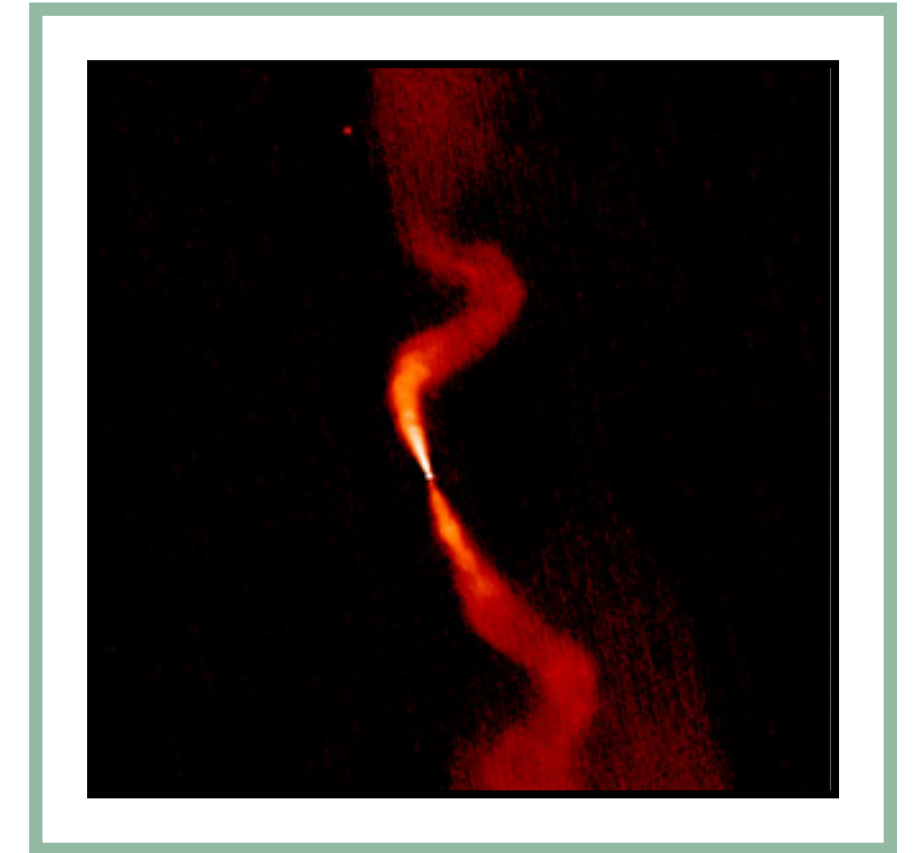
$$\mathcal{L}(F) : \mathbb{R}^d \rightarrow \mathbb{R}^{c_{\text{out}}}$$

i.e. at each pixel it takes a set of input features with dimension c_{in} and produces a set of output features with dimension c_{out} .

The function is **equivariant** to translations $t \in (\mathbb{R}^d, +)$ if

$$\mathcal{L}(t \triangleright F) = t \triangleright \mathcal{L}(F)$$

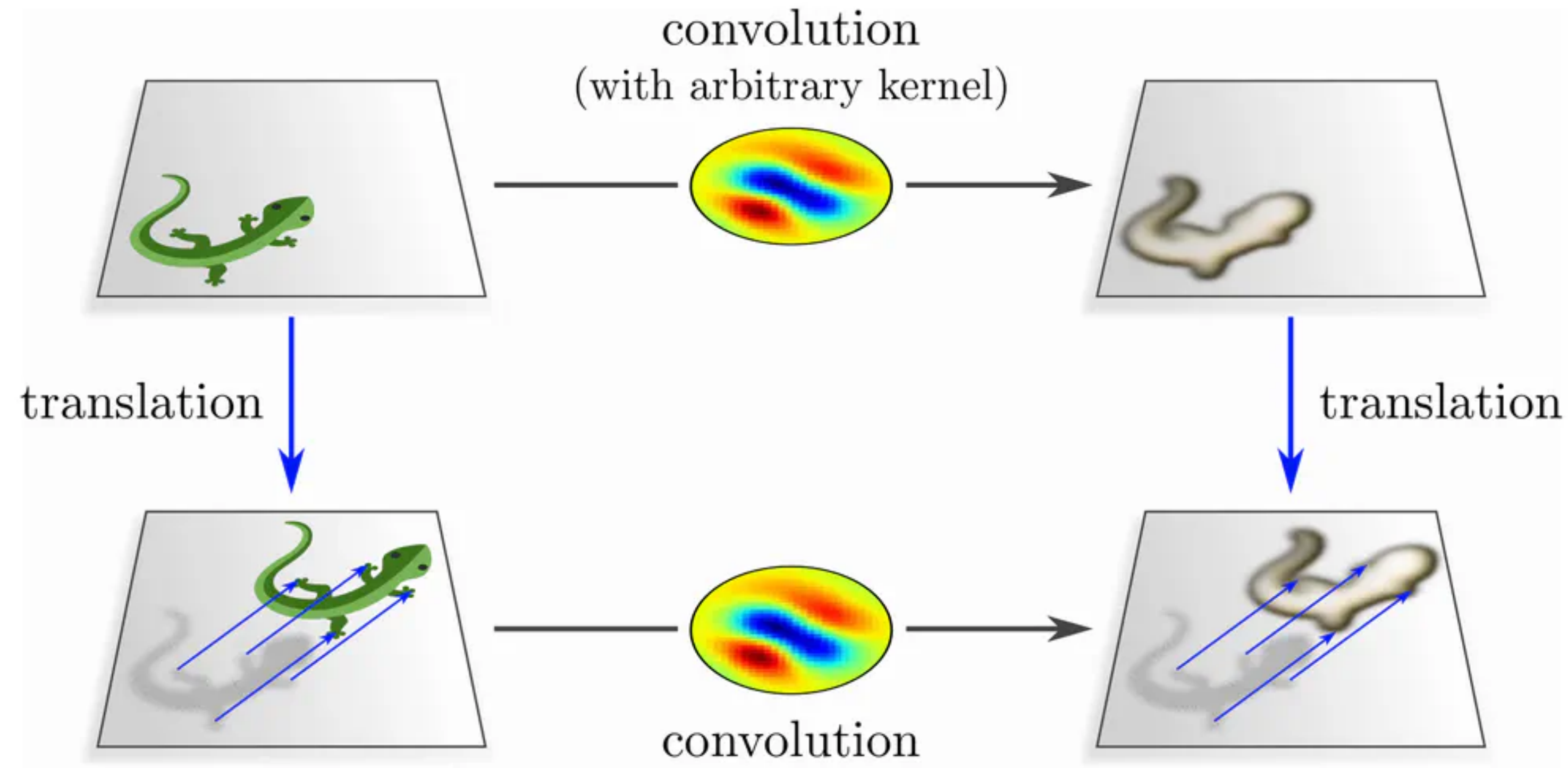
i.e. it commutes with the actions of the group.



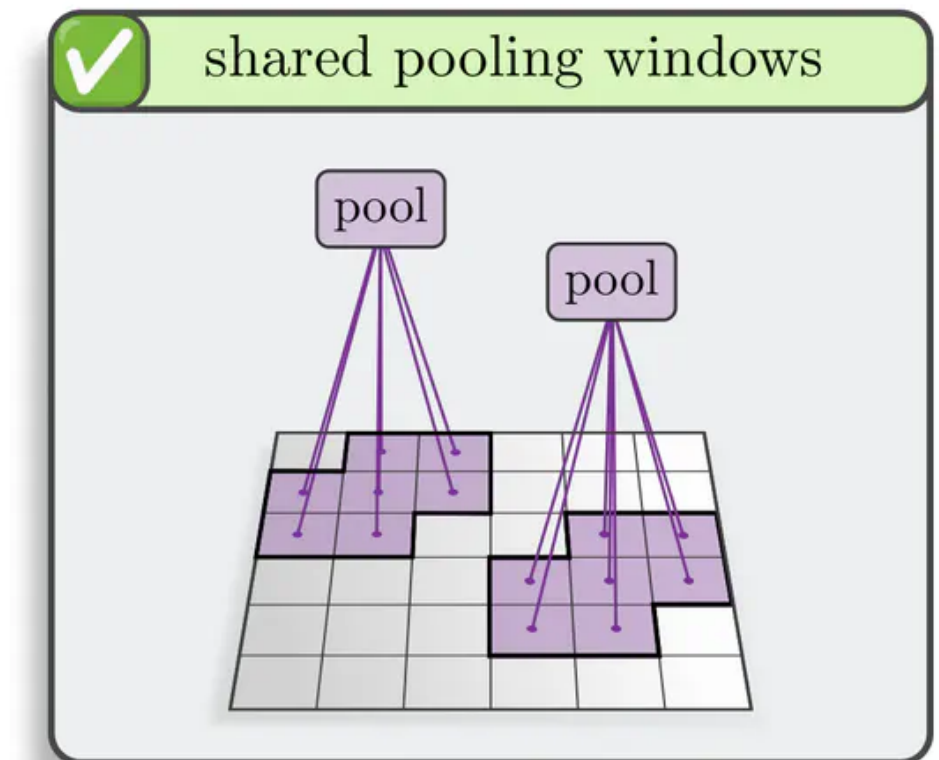
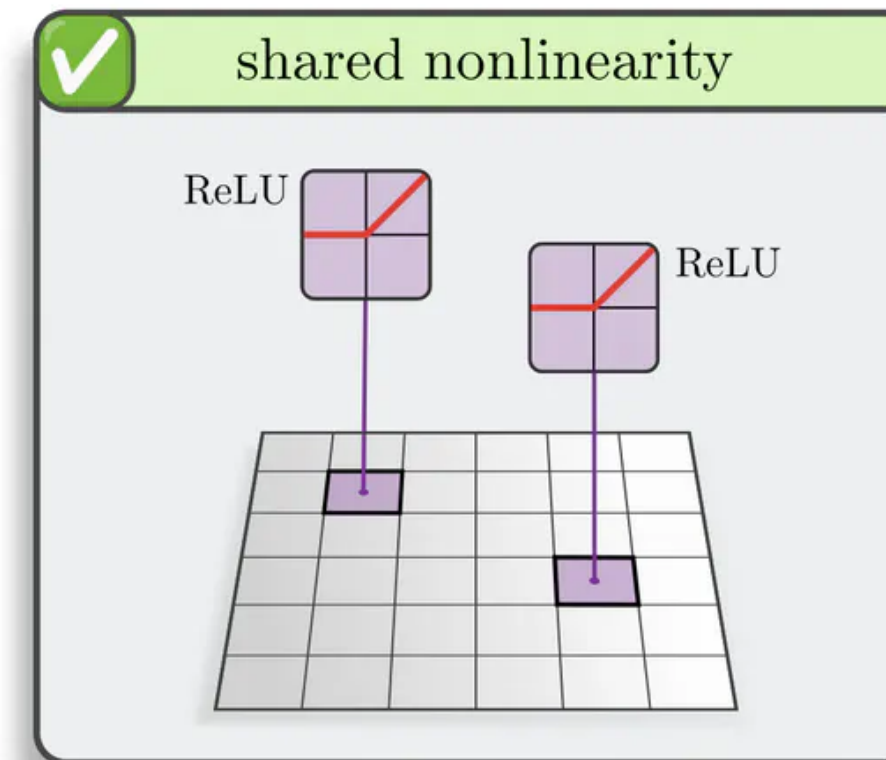
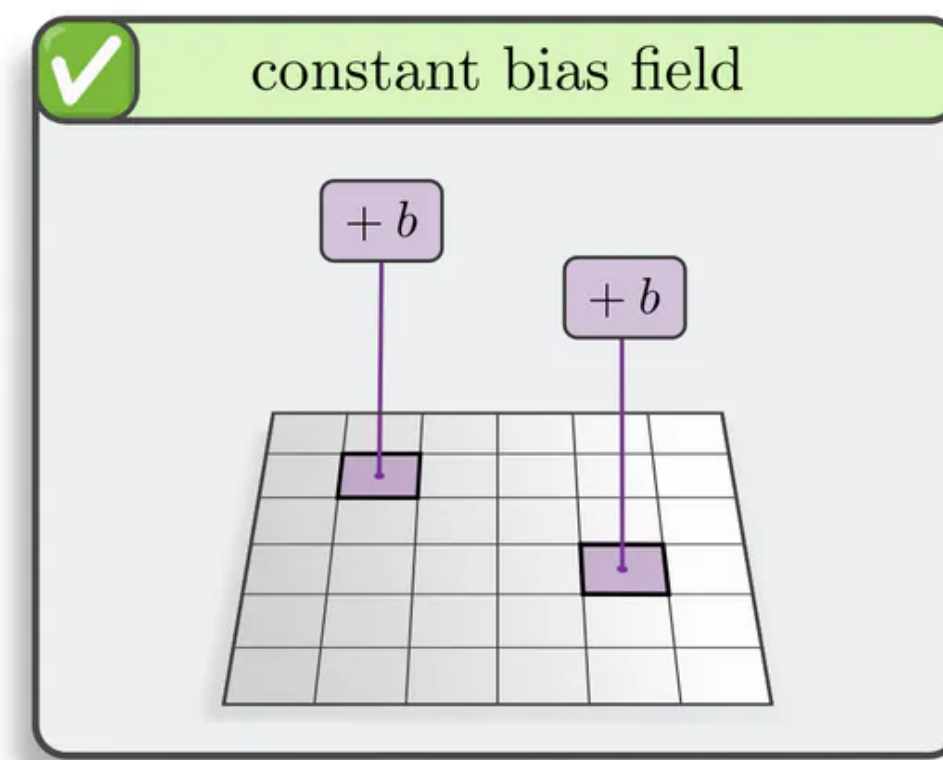
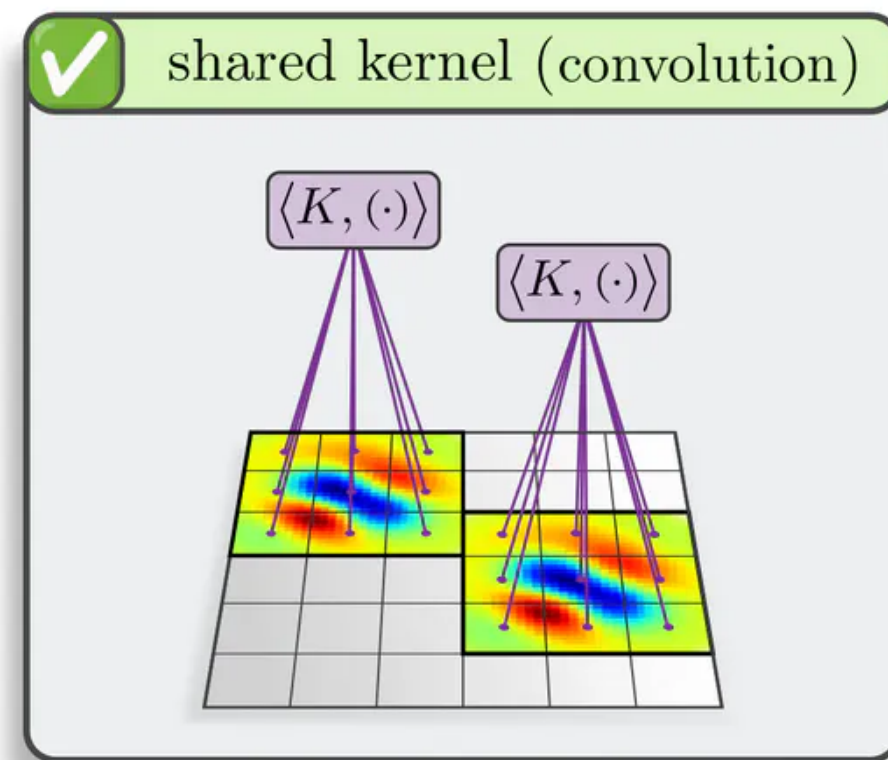
spatial weight sharing



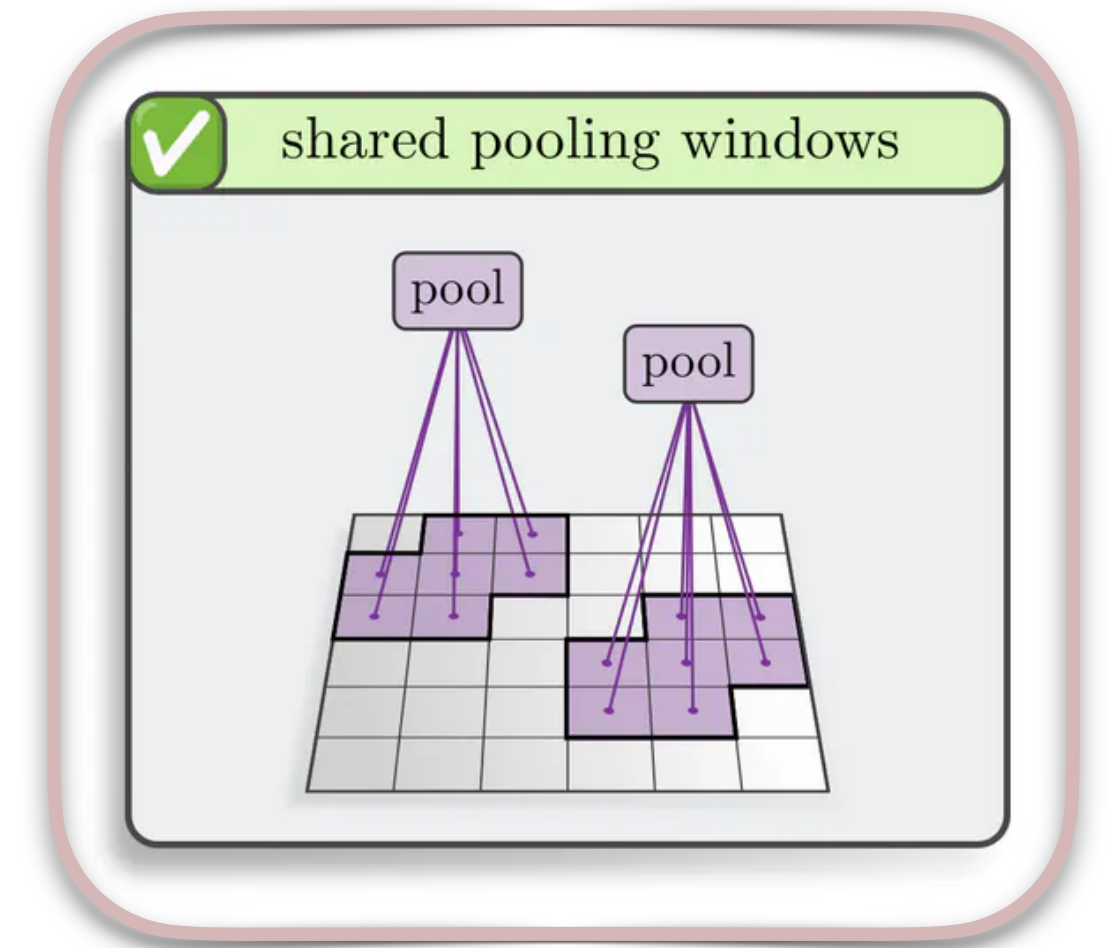
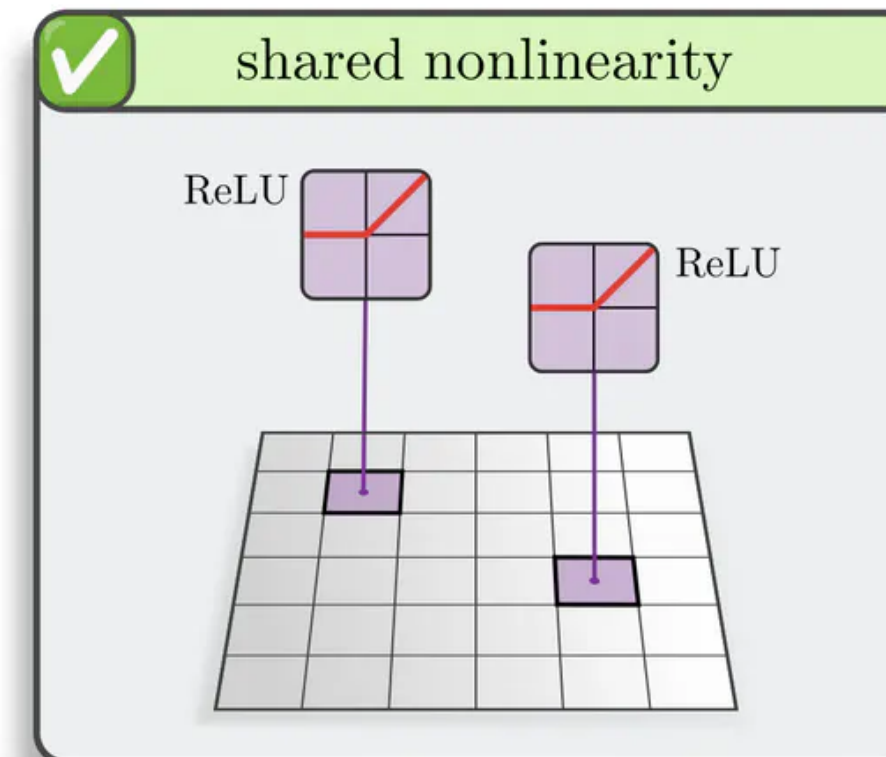
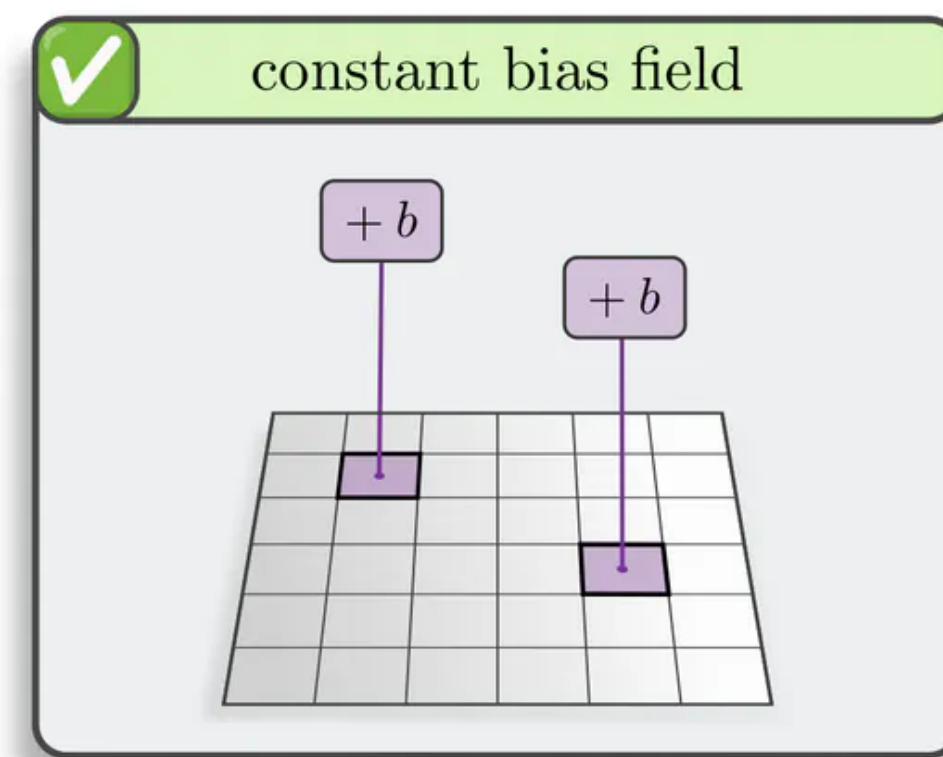
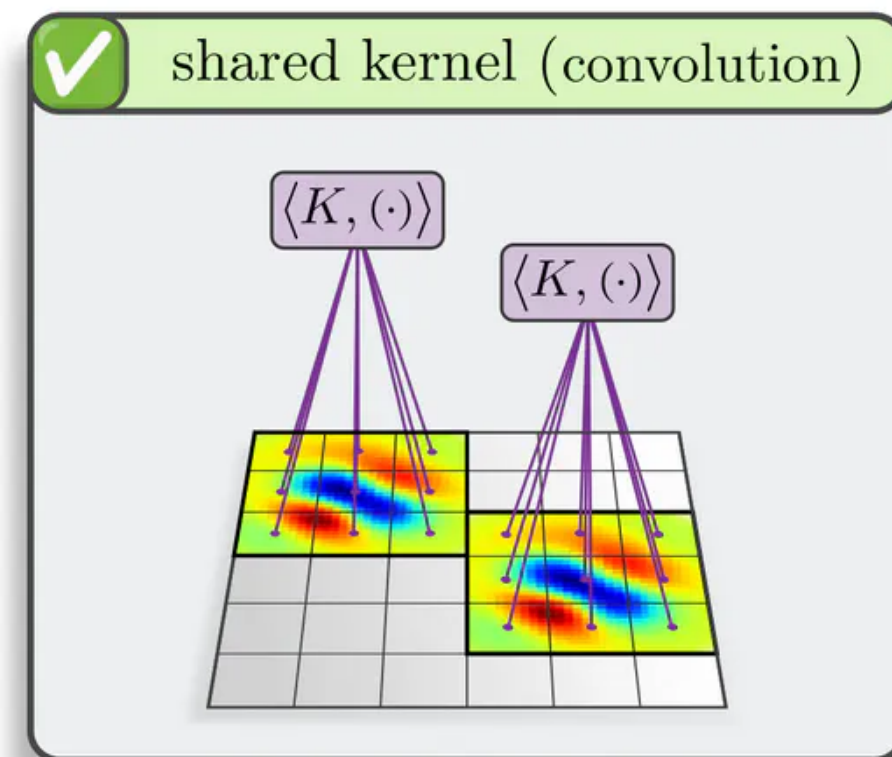
translational equivariance



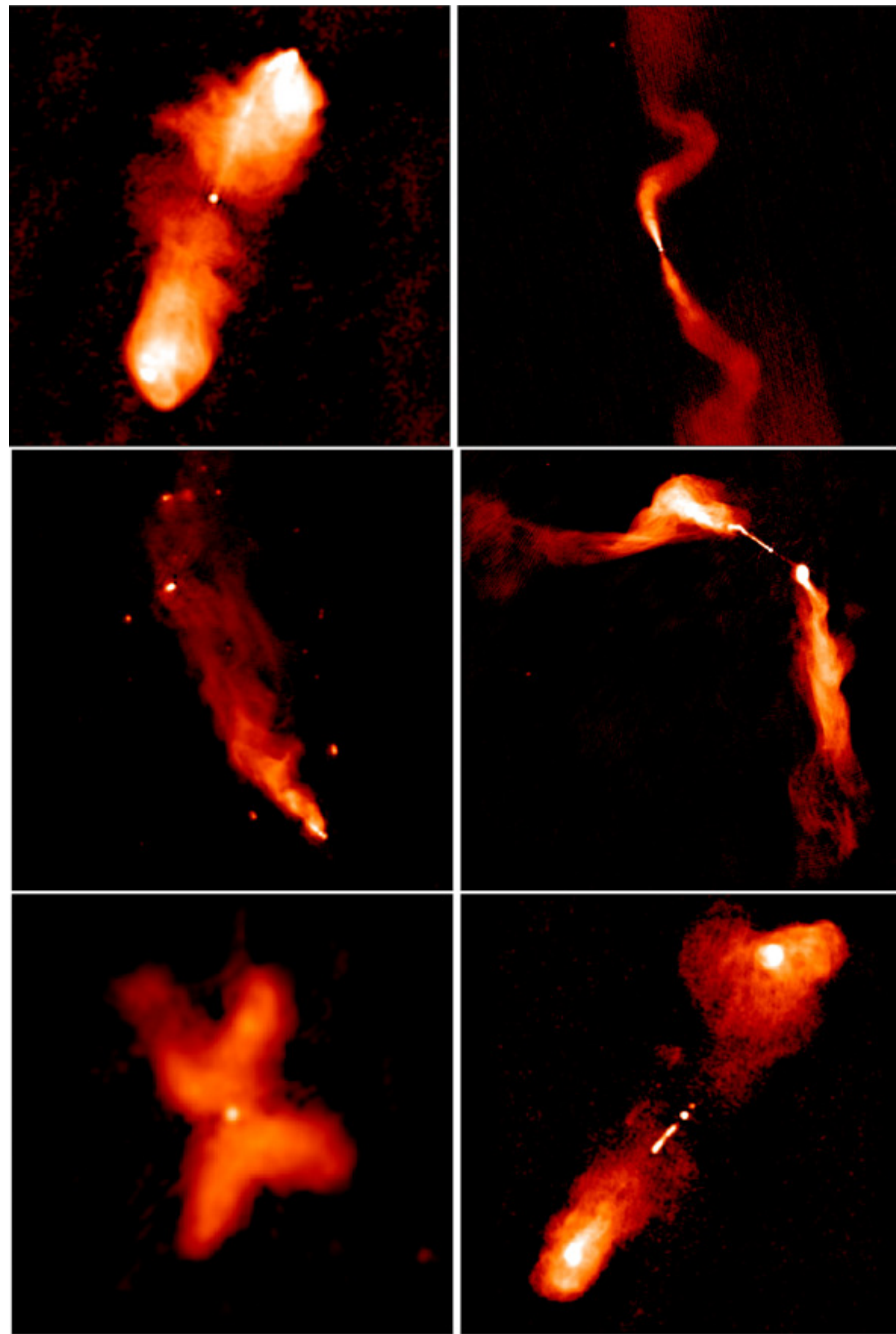
Notes on keeping convolutions equivariant:



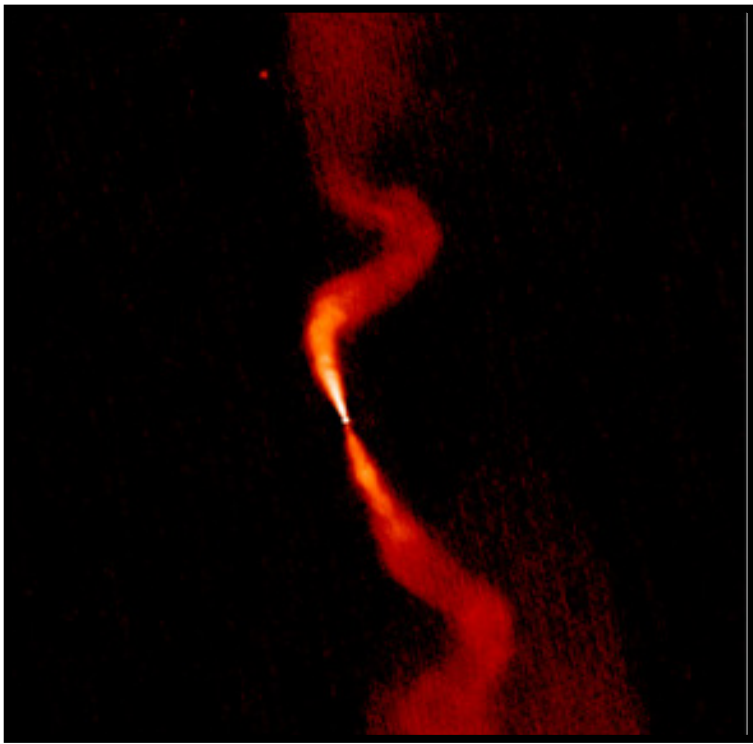
[All of these things are generally done as standard in CNNs]



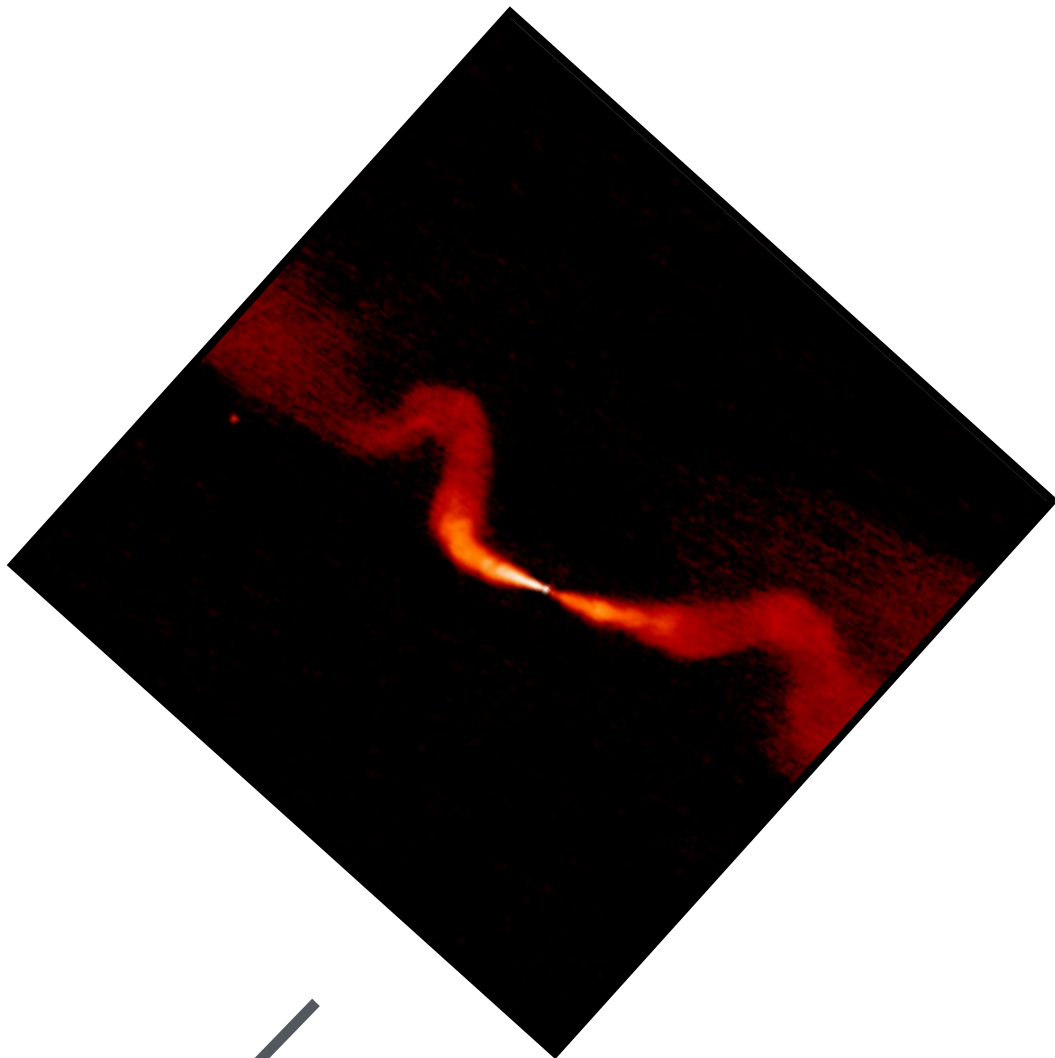
Both *max* and *average* (global) pooling operations result in translationally **invariant outputs**.



invariance



rotate

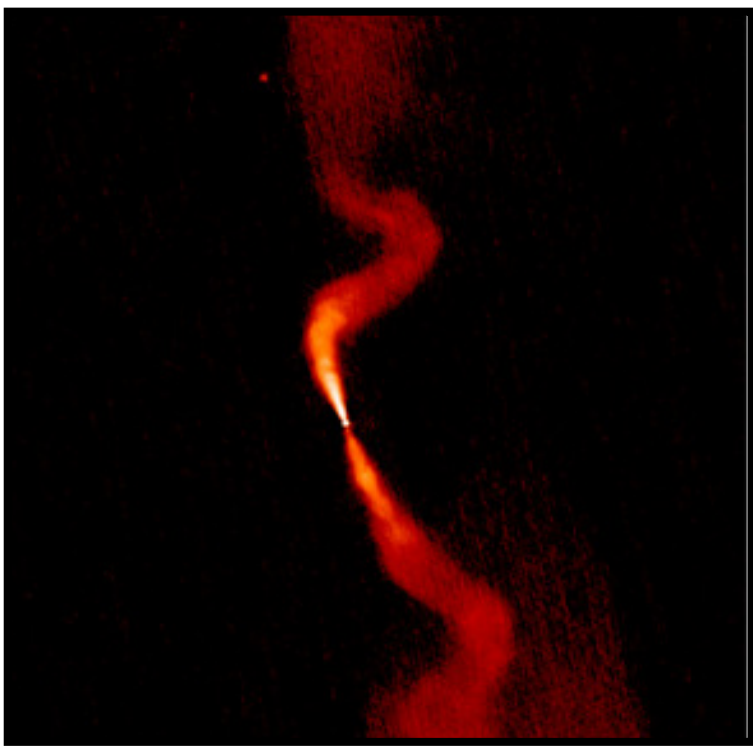
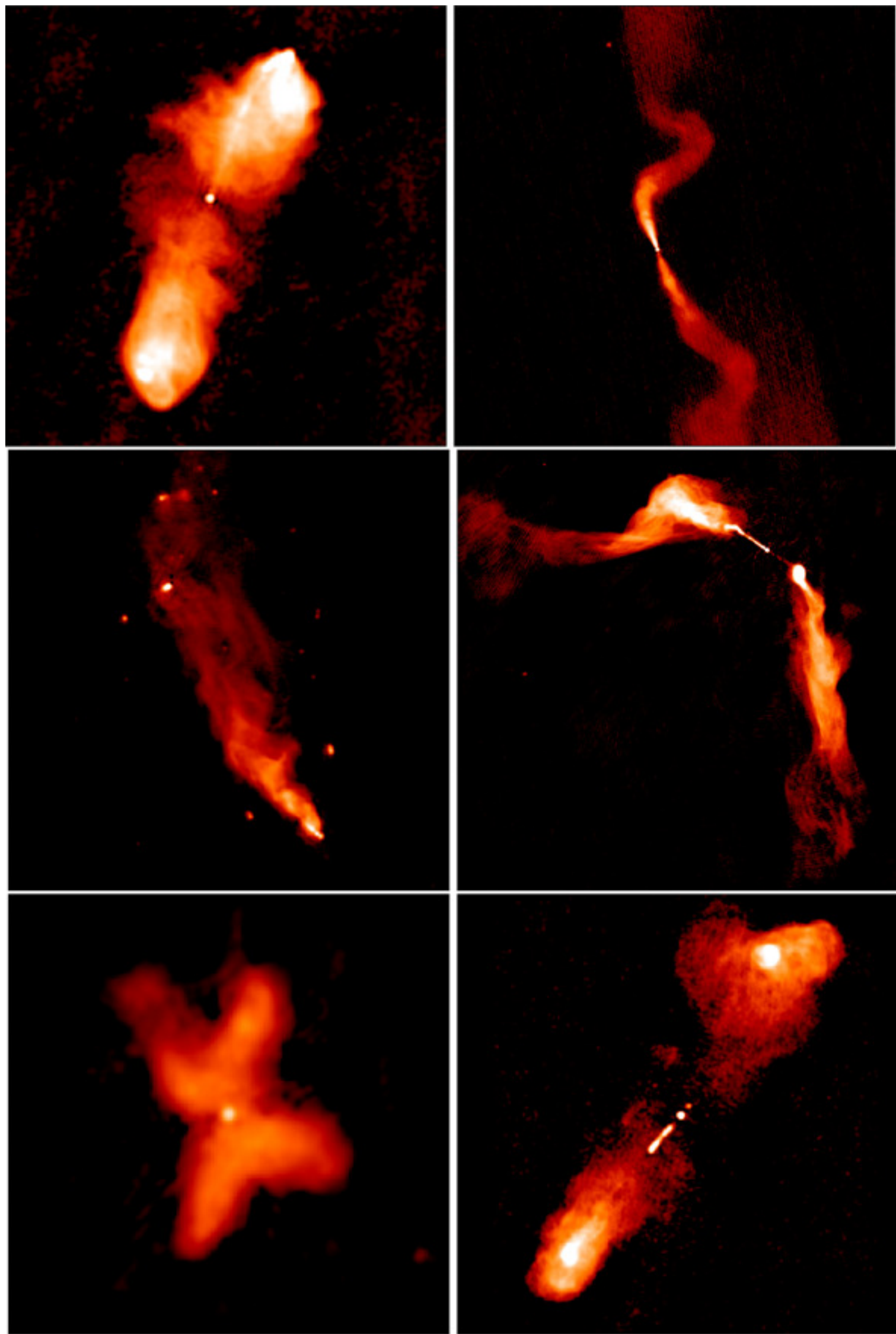


class

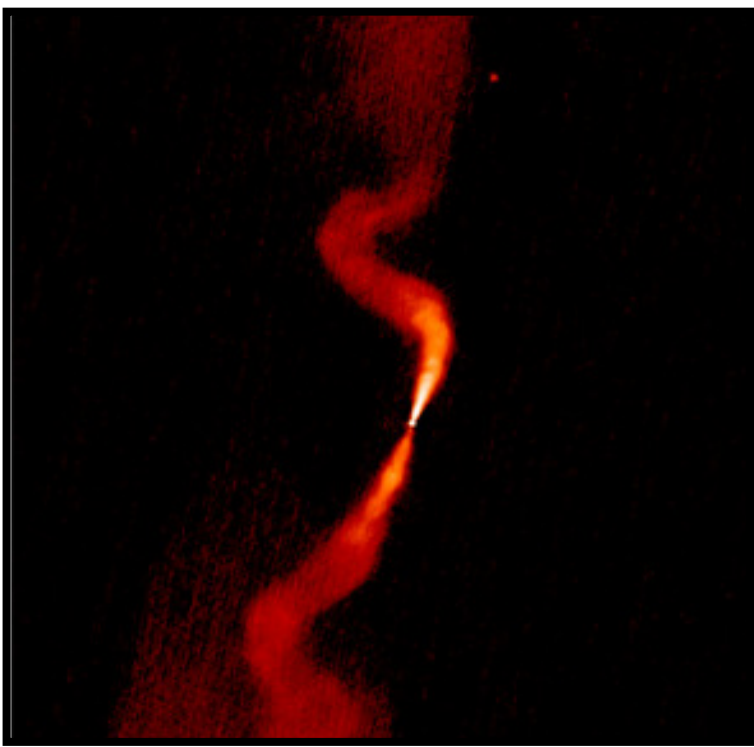
class

FR I

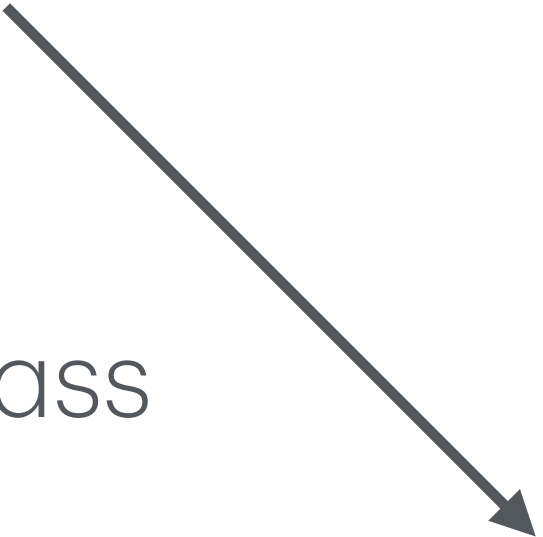
invariance



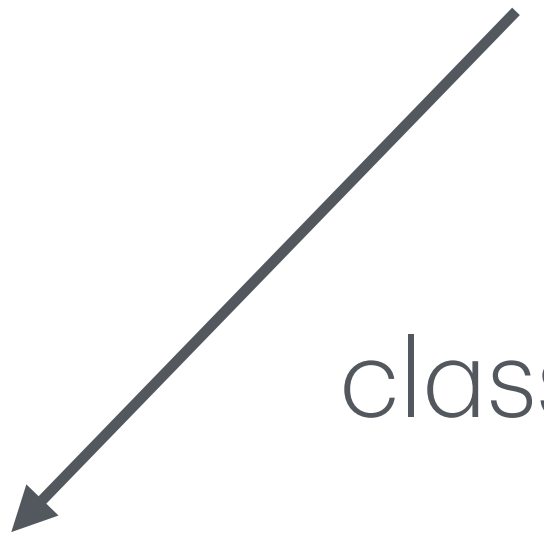
reflect



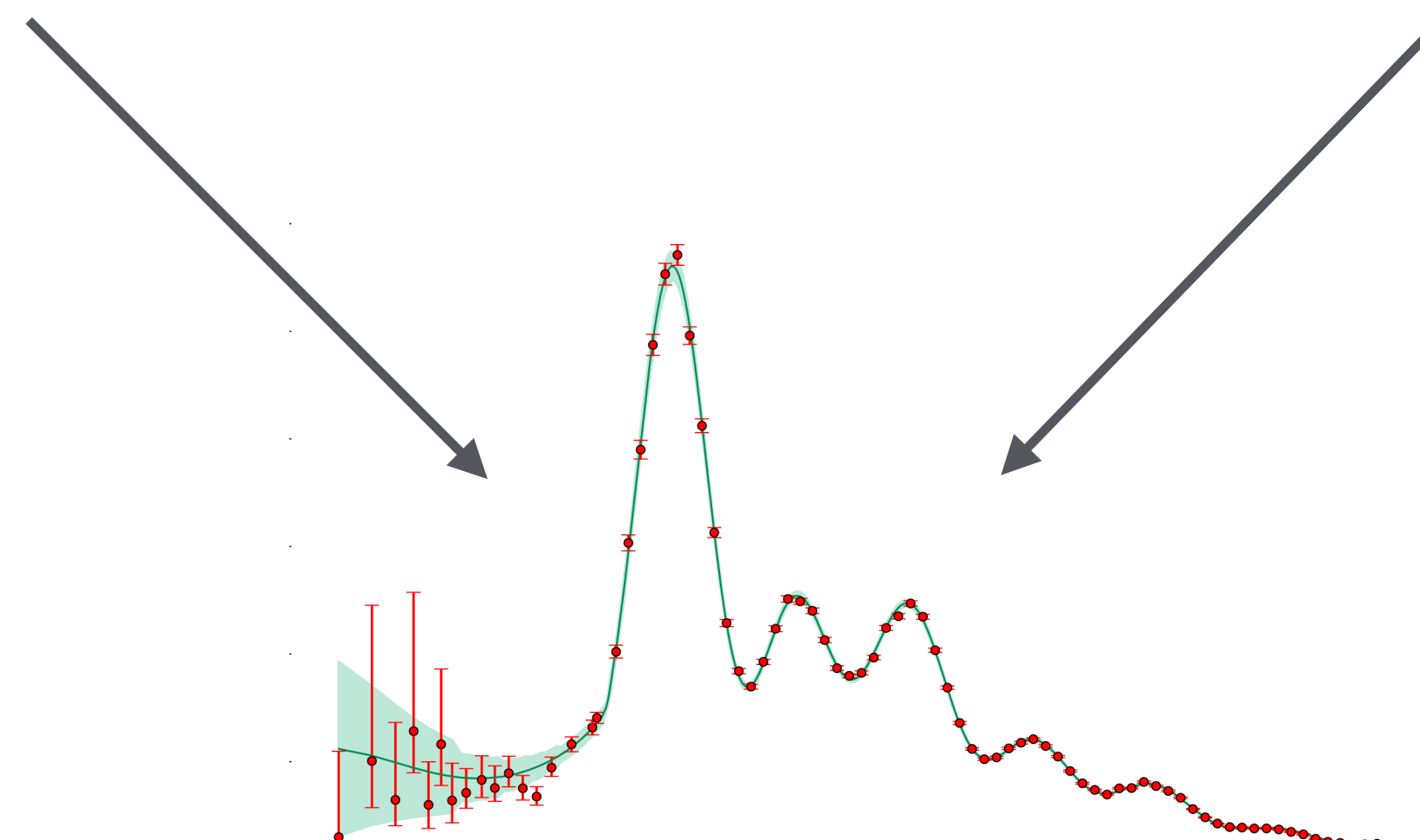
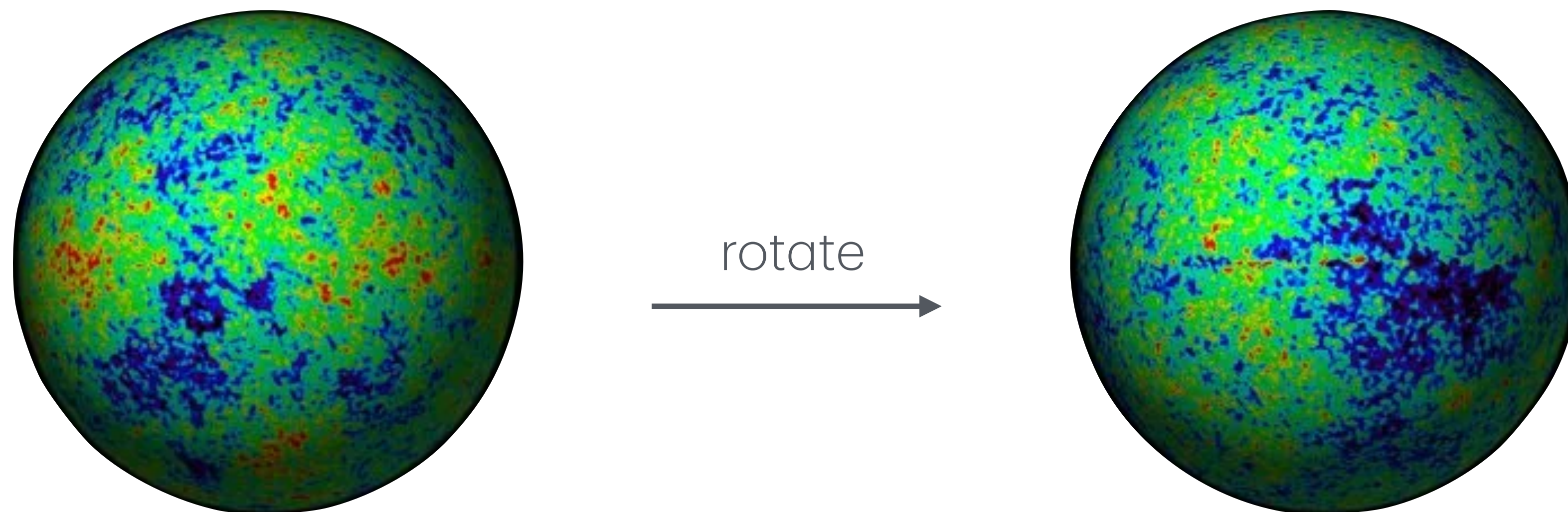
class



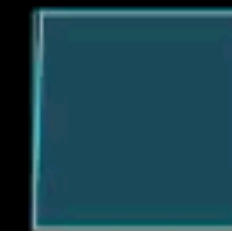
class



FR I

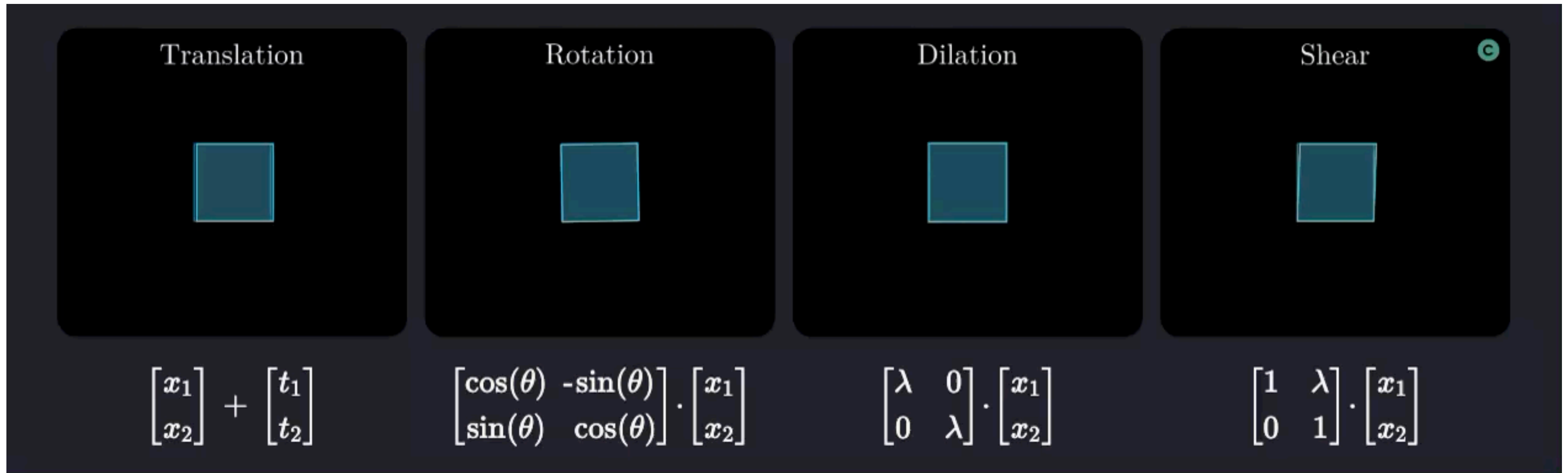


translations are special...

Translation	Rotation	Dilation	Shear
			
$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$	$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$	$\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$	$\begin{bmatrix} 1 & \lambda \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

summation

matrix multiplication



$$(\mathbb{R}^d, +)$$

$$G \leq GL(d)$$

Transformations that require matrix multiplication are described by the general linear group, $GL(d)$, but normally we only need a specific subset of these transformations:

$$G \leq GL(d)$$

For example, if we only need rotations and reflections, we would use the $G = O(d)$ subgroup.

The combination of translations and specific matrix multiplications are called affine groups:

$$Aff(G) := (\mathbb{R}^d, +) \rtimes G$$

For example, in 2d Euclidean space, we might have:

$$E(2) = (\mathbb{R}^2, +) \rtimes O(2)$$

which is the group containing translations, rotations and reflections for flat images.

translational weight sharing

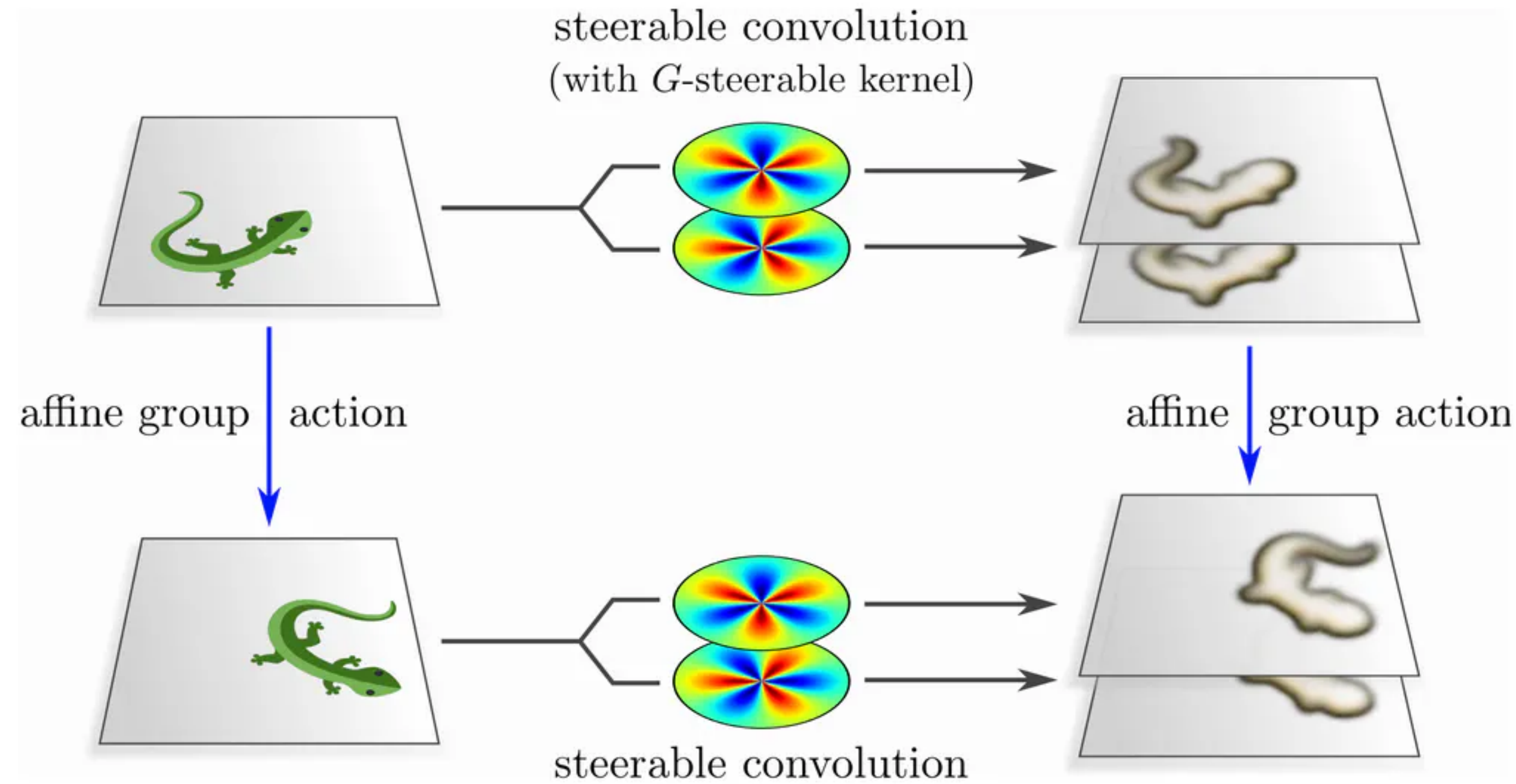


translational equivariance

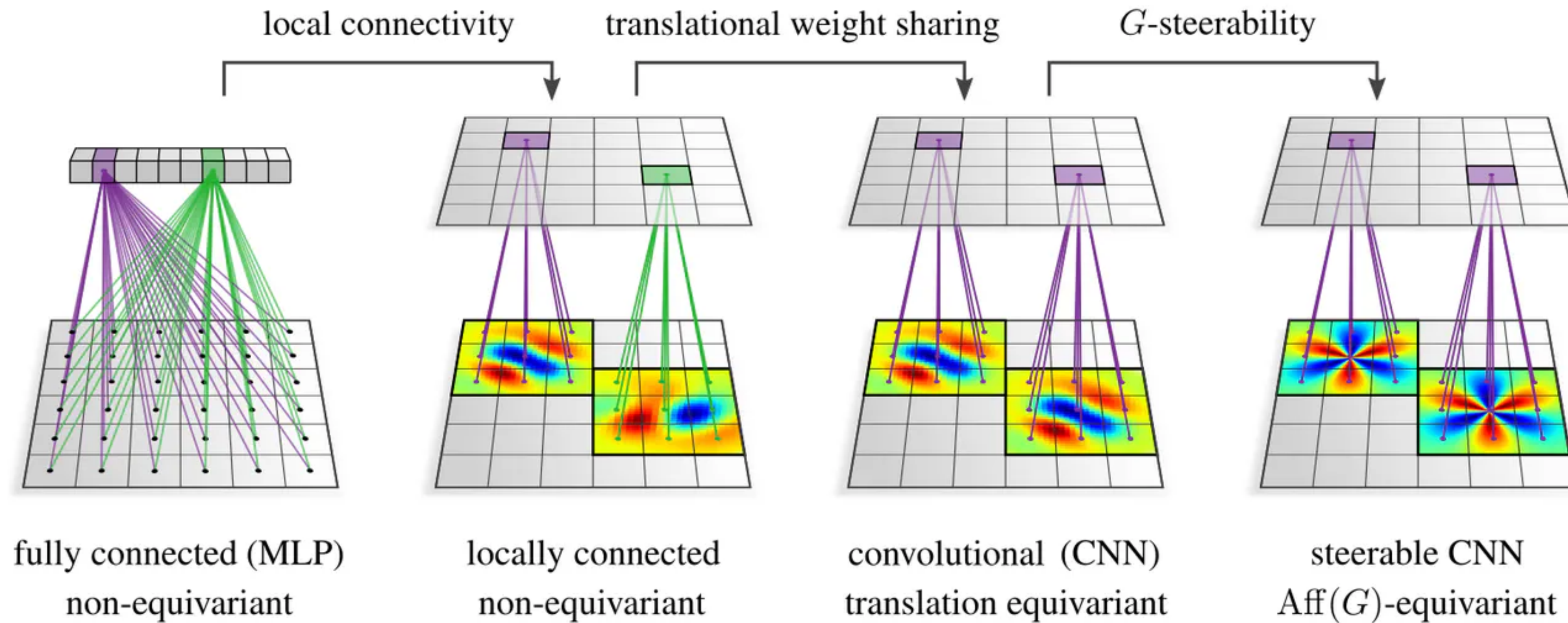
affine weight sharing

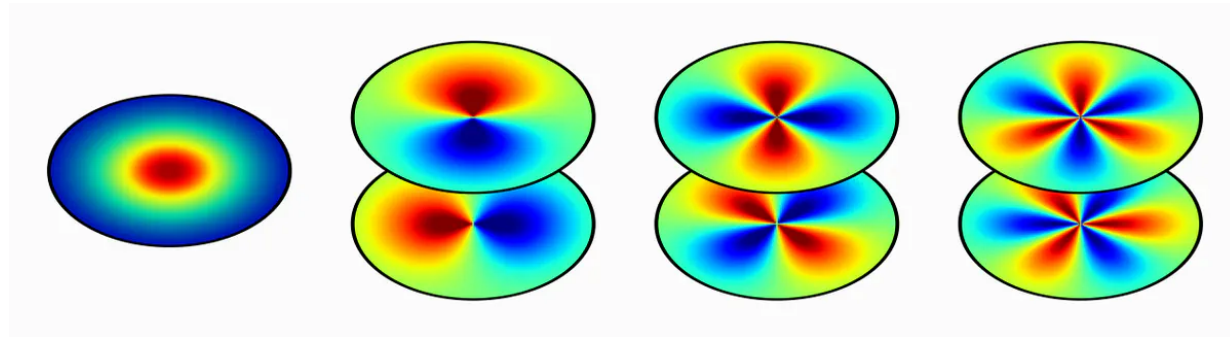


affine equivariance



additional weight sharing over
 G -actions: " G -steerability"

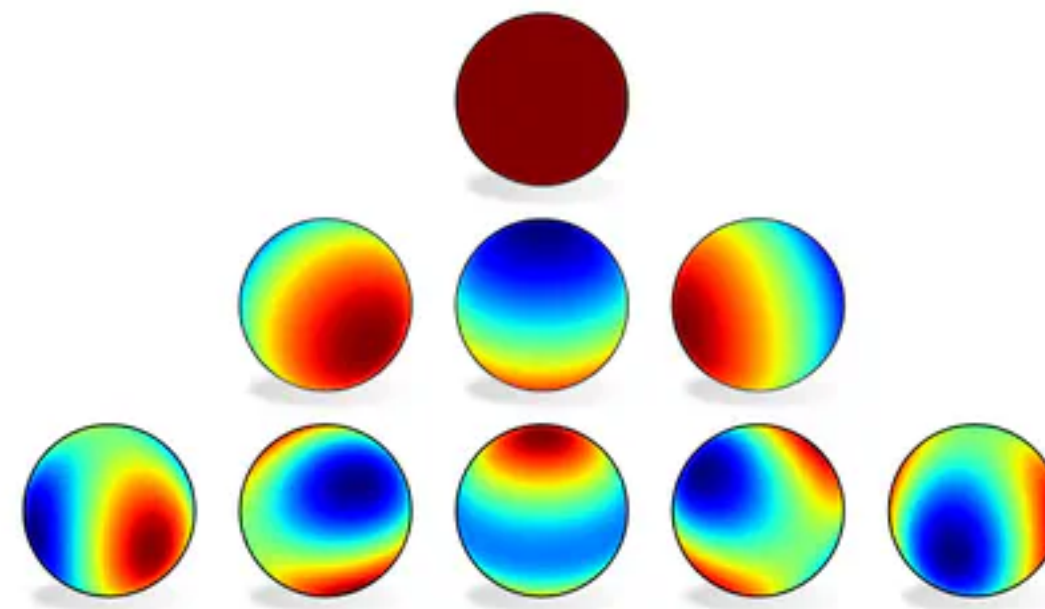




Harmonic networks

$SE(2)$ -equivariant
networks using harmonic
kernels.

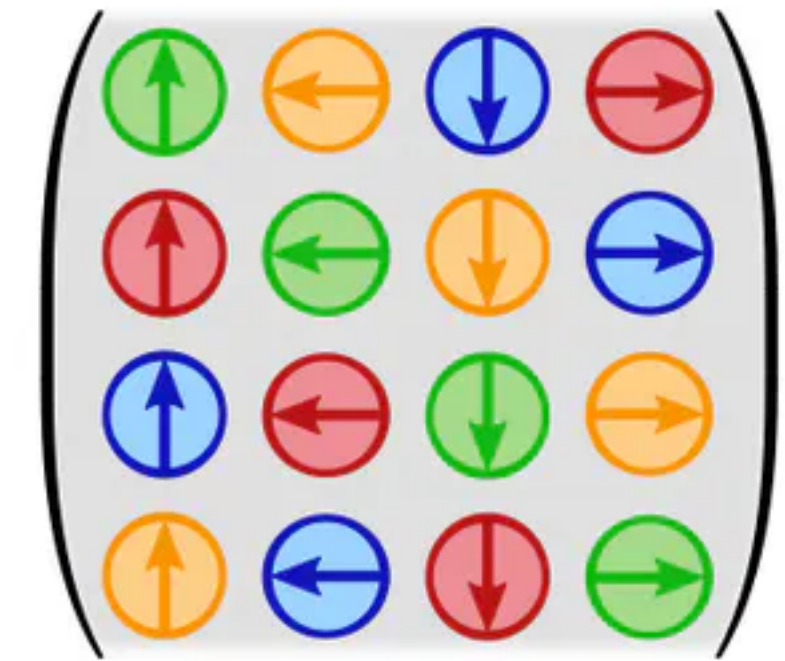
Worrall+ 2016
arXiv:1612.04642



Tensor field networks

$E(3)$ -equivariant
networks using spherical
harmonics + Clebsh-
Gordon decomposition

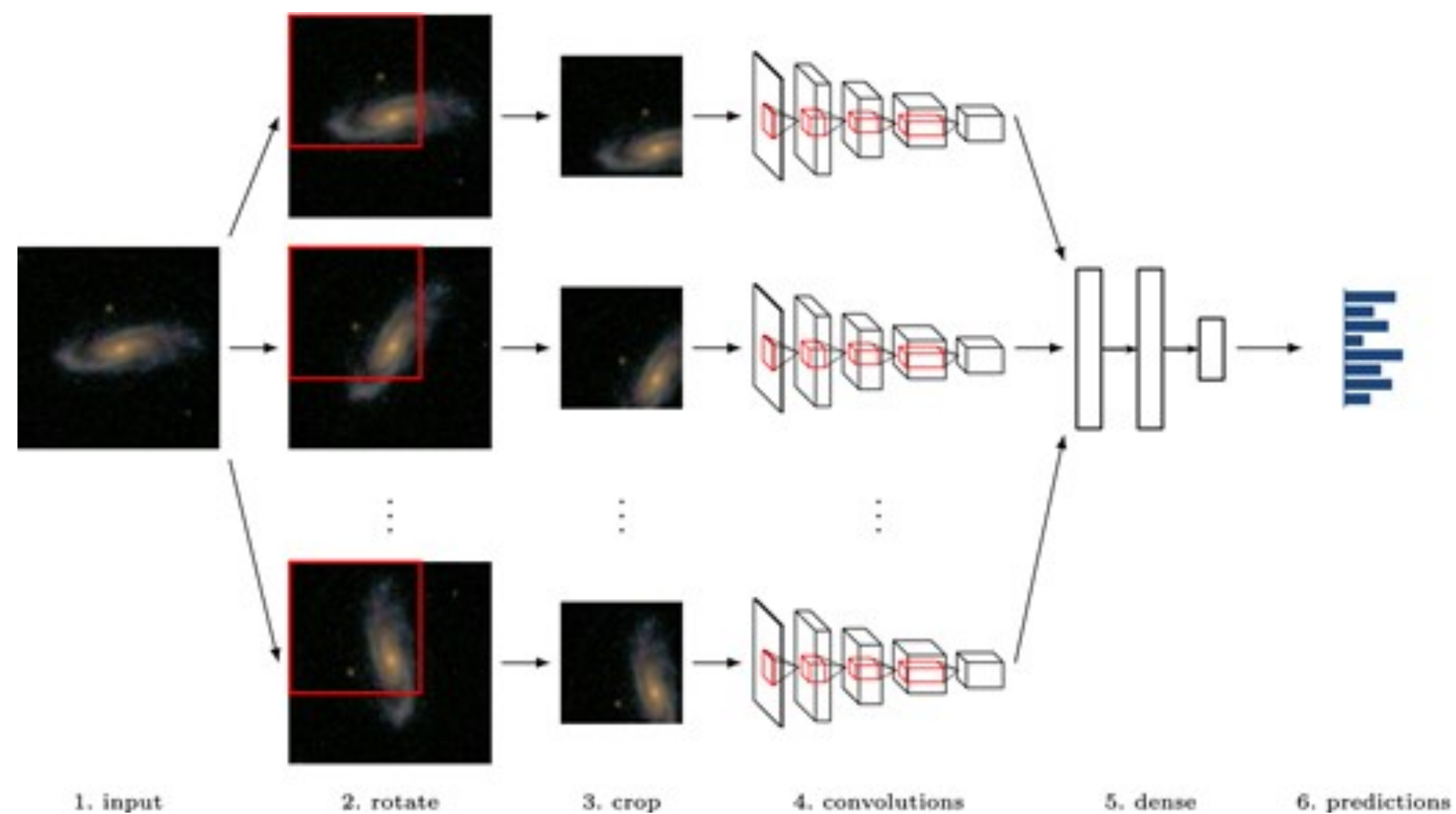
Thomas+ 2018
arXiv:1802.08219



Group equivariant CNNs

G -equivariant networks
using discrete groups

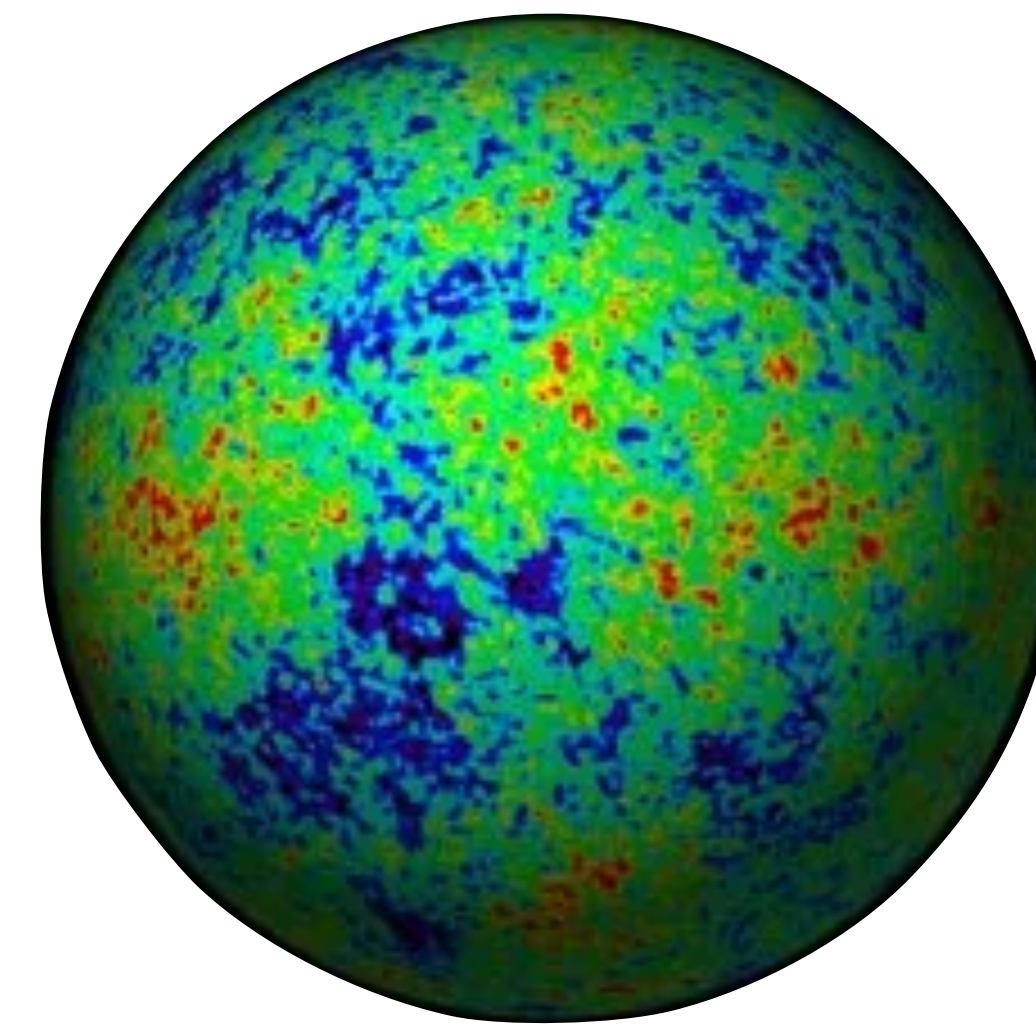
Cohen & Welling+ 2016
arXiv:1602.07576



Rotation-invariant classification

Multi-branch feature
extraction

Dieleman+ 2015
arXiv:1503.07077



Spherical CNNs

Scattering networks on
the sphere

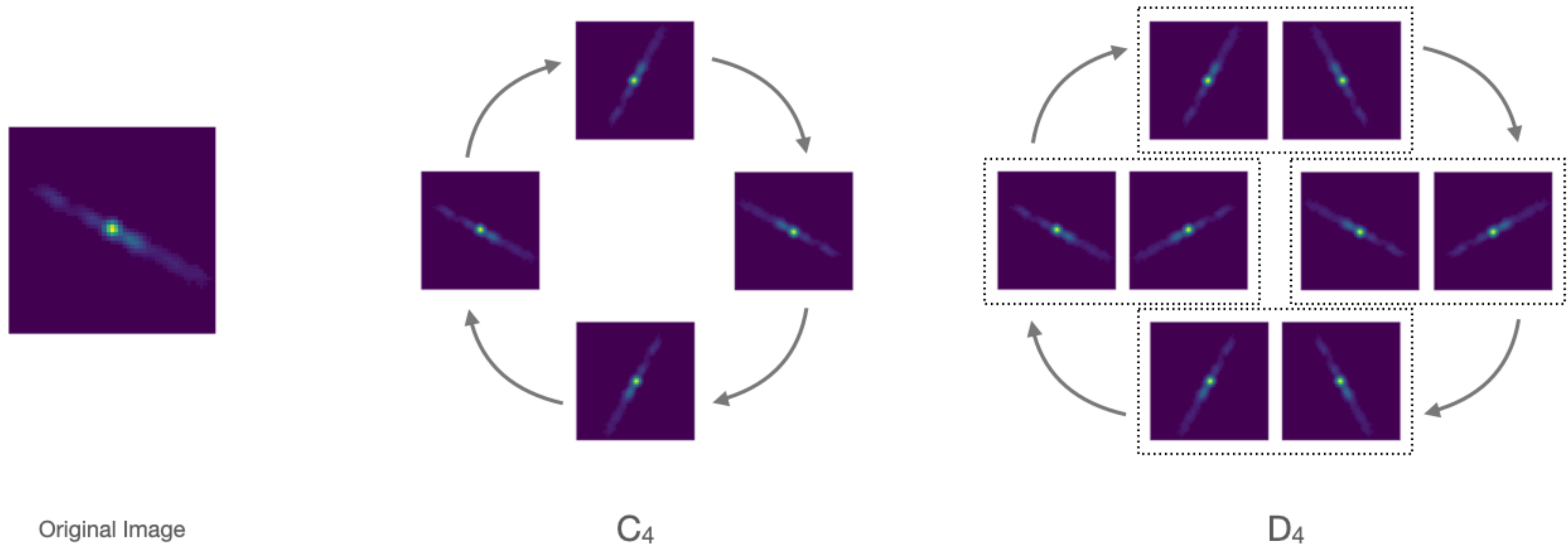
McEwen+ 2021
arXiv:2102.02828

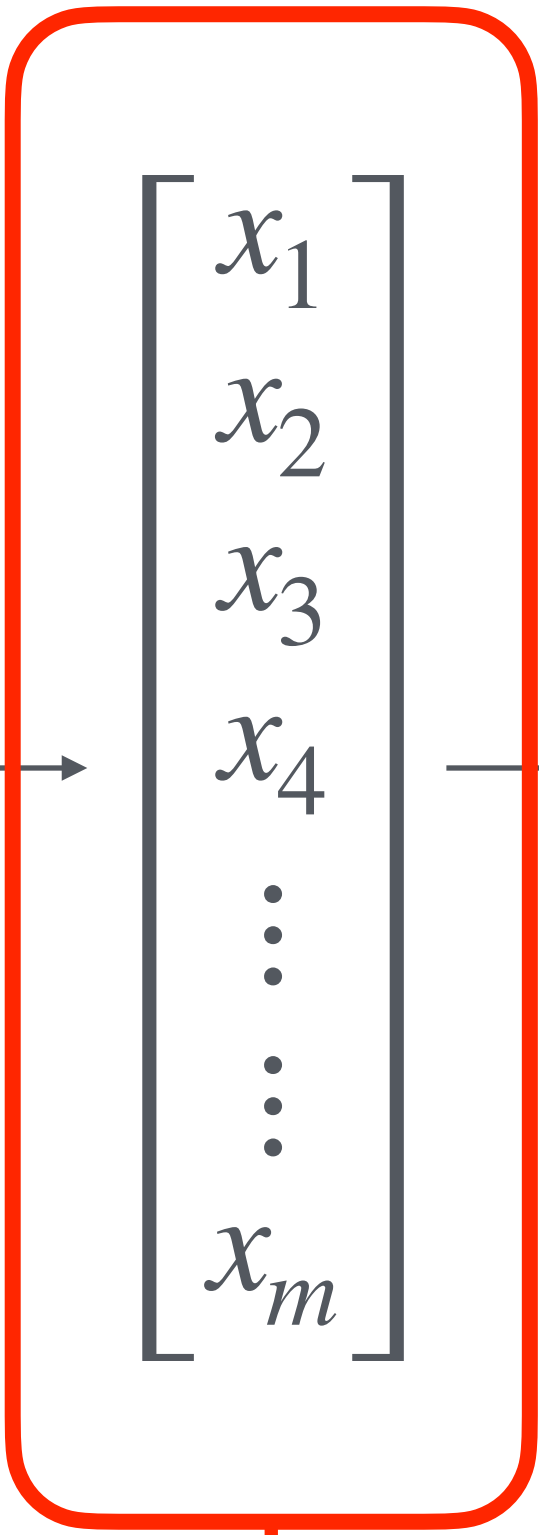
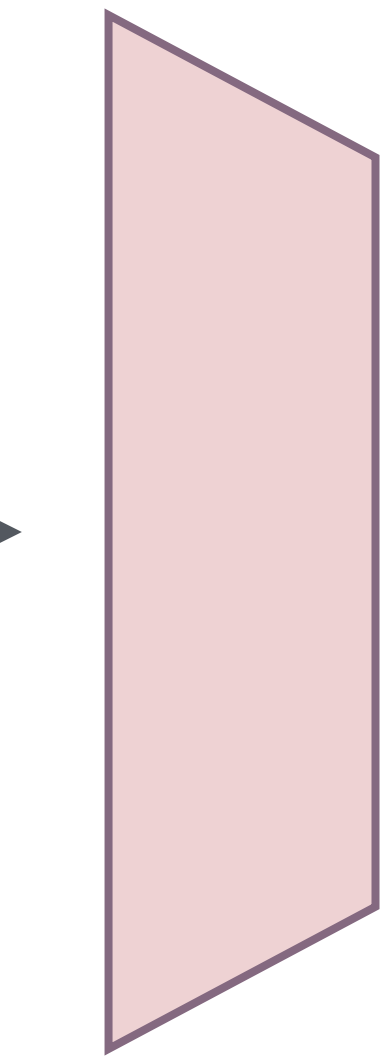
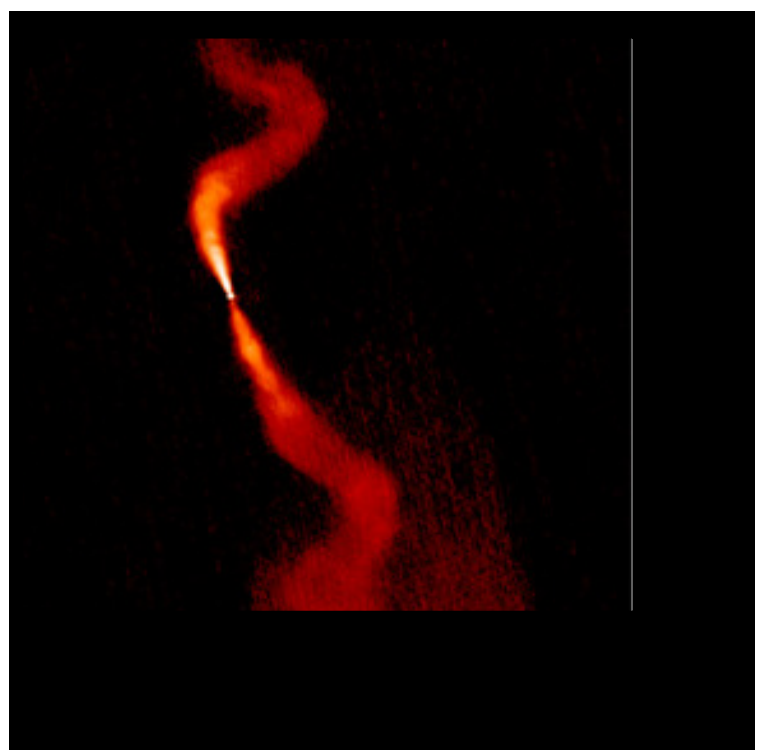
	space	matrix group G	global symmetry $\text{Aff}(G)$	representation ρ	citation
1	$\mathbb{R}^d$	$\{e\}$	$(\mathbb{R}^d, +)$	trivial	conventional CNNs [175, 348]
2		scaling $\mathcal{S}$	$(\mathbb{R}^1, +) \rtimes \mathcal{S}$	regular	[248]
3	$\mathbb{R}^1$	reflection $\mathcal{R}$	$(\mathbb{R}^1, +) \rtimes \mathcal{R}$	regular	[193]
4				irreps	[193]
5		reflection $\mathcal{R}$	$(\mathbb{R}^2, +) \rtimes \mathcal{R}$	regular	[322]
6				irreps	[335, 322]
7		SO(2)	SE(2)	regular	[71, 52, 358, 53, 324, 12, 125, 258, 275, 27, 70, 76] [322, 110, 170, 279, 317, 247, 215, 276, 277, 37, 23] [270, 10, 91, 306, 113, 216, 311, 122, 223, 43, 116]
8				quotients	[53, 322]
9				regular $\xrightarrow{\text{pool}}$ trivial	[52, 195, 322]
10	$\mathbb{R}^2$			regular $\xrightarrow{\text{pool}}$ vector	[196, 322]
11				trivial	[144, 322]
12				irreps	[322]
13		O(2)	E(2)	regular	[71, 52, 125, 53, 322] [216, 110, 270, 23]
14				quotients	[53]
15				regular $\xrightarrow{\text{pool}}$ trivial	[322]
16				induced SO(2)-irreps	[322]
17		scaling $\mathcal{S}$	$(\mathbb{R}^2, +) \rtimes \mathcal{S}$	regular	[334, 281, 10, 359]
18				regular $\xrightarrow{\text{pool}}$ trivial	[107]
19		SO(2) $\times$ $\mathcal{S}$	$(\mathbb{R}^2, +) \rtimes (\text{SO}(2) \times \mathcal{S})$	regular	[349]
20				irreps	[323, 301, 211, 161, 3, 184]
21		SO(3)	SE(3)	quaternion	[345]
22				regular	[91, 329, 333]
23				regular $\xrightarrow{\text{pool}}$ trivial	[4]
24	$\mathbb{R}^3$			irreps	[8]
25		O(3)	E(3)	regular	[329]
26				quotient O(3)/O(2)	[136]
27				irrep $\xrightarrow{\text{norm}}$ trivial	[233]
28		C ₄	$(\mathbb{R}^3, +) \rtimes \text{C}_4$	regular	[289]
29		D ₄	$(\mathbb{R}^3, +) \rtimes \text{D}_4$	regular	[289]
30	Minkowski	SO($d-1, 1$)	$(\mathbb{R}^d, +) \rtimes \text{SO}(d-1, 1)$	irreps	[274]

To make the G-steerability numerically tractable, we can take discrete realisations of different groups.

For example:

- the set of 90 degree rotations: $C_4 < G = SO(2)$
- The set of 90 degree rotations and reflections: $D_4 < G = O(2)$

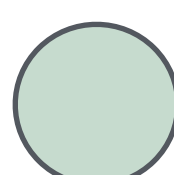
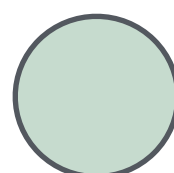
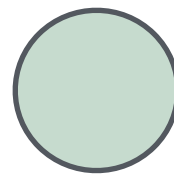
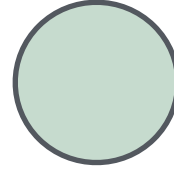
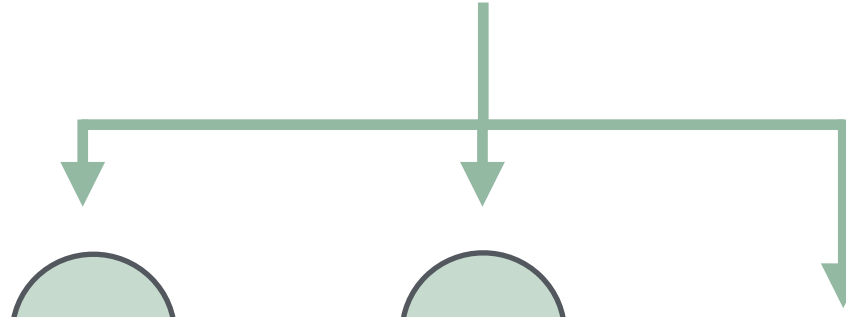
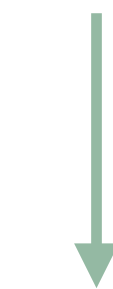




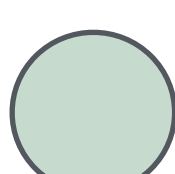
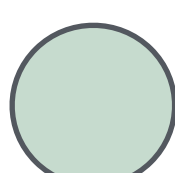
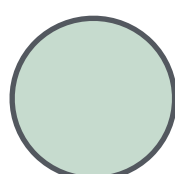
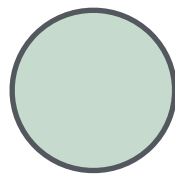
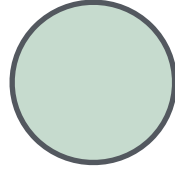
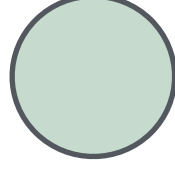
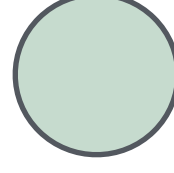
input layer

hidden layers

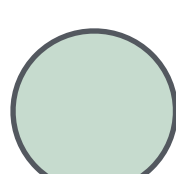
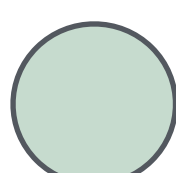
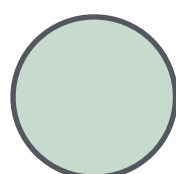
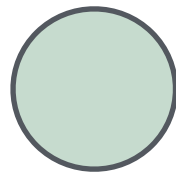
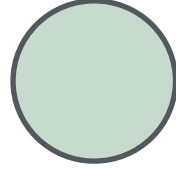
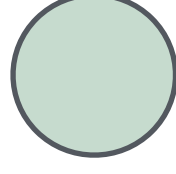
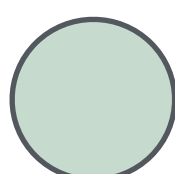
output layer



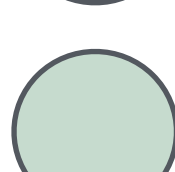
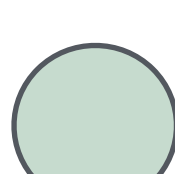
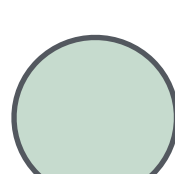
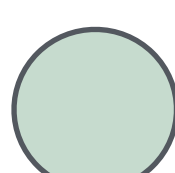
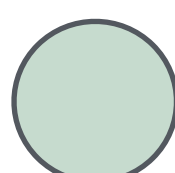
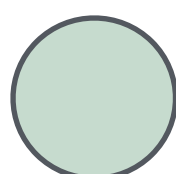
⋮



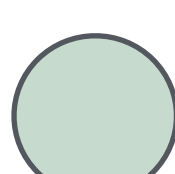
⋮



⋮



⋮



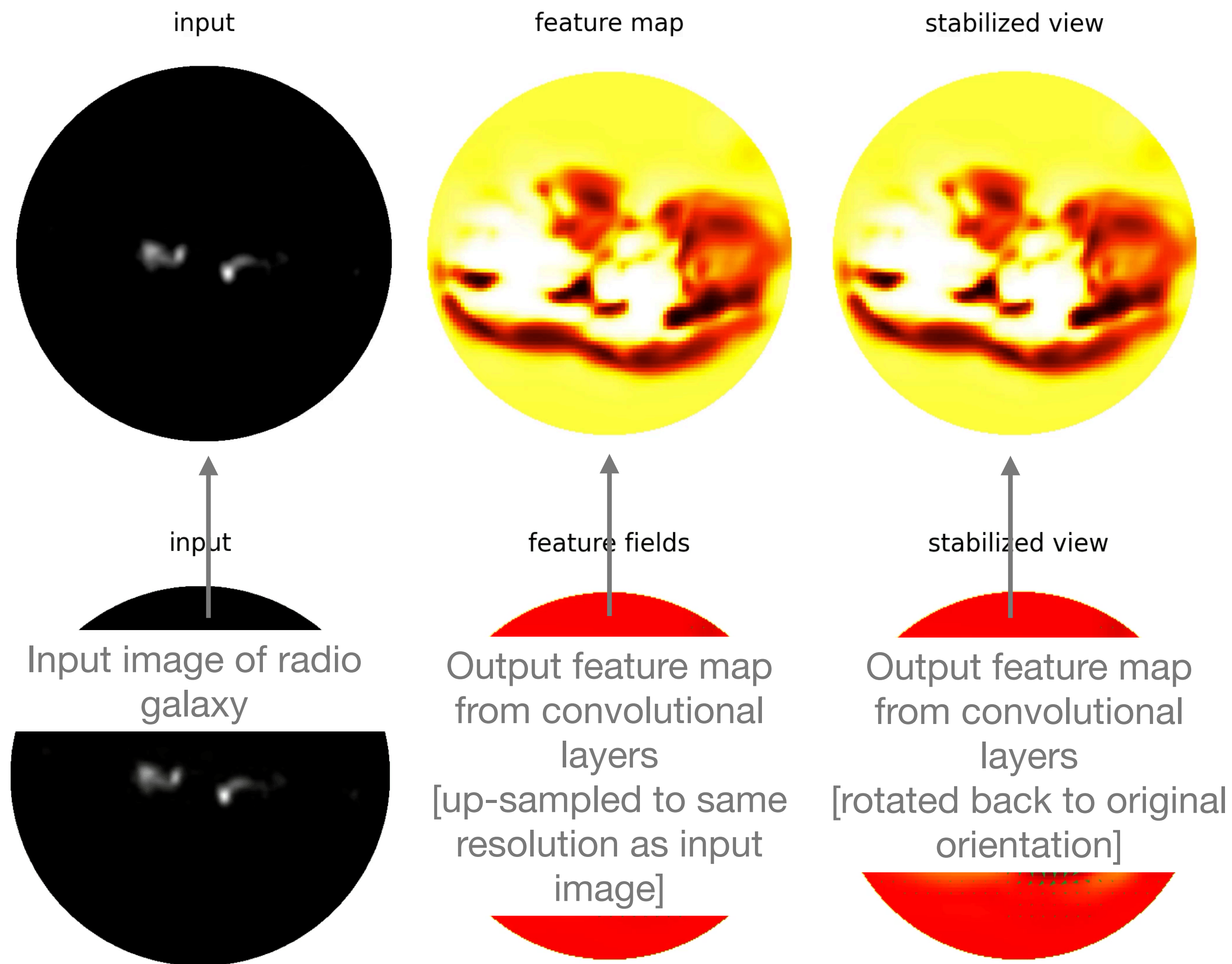
$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

features

convolutions

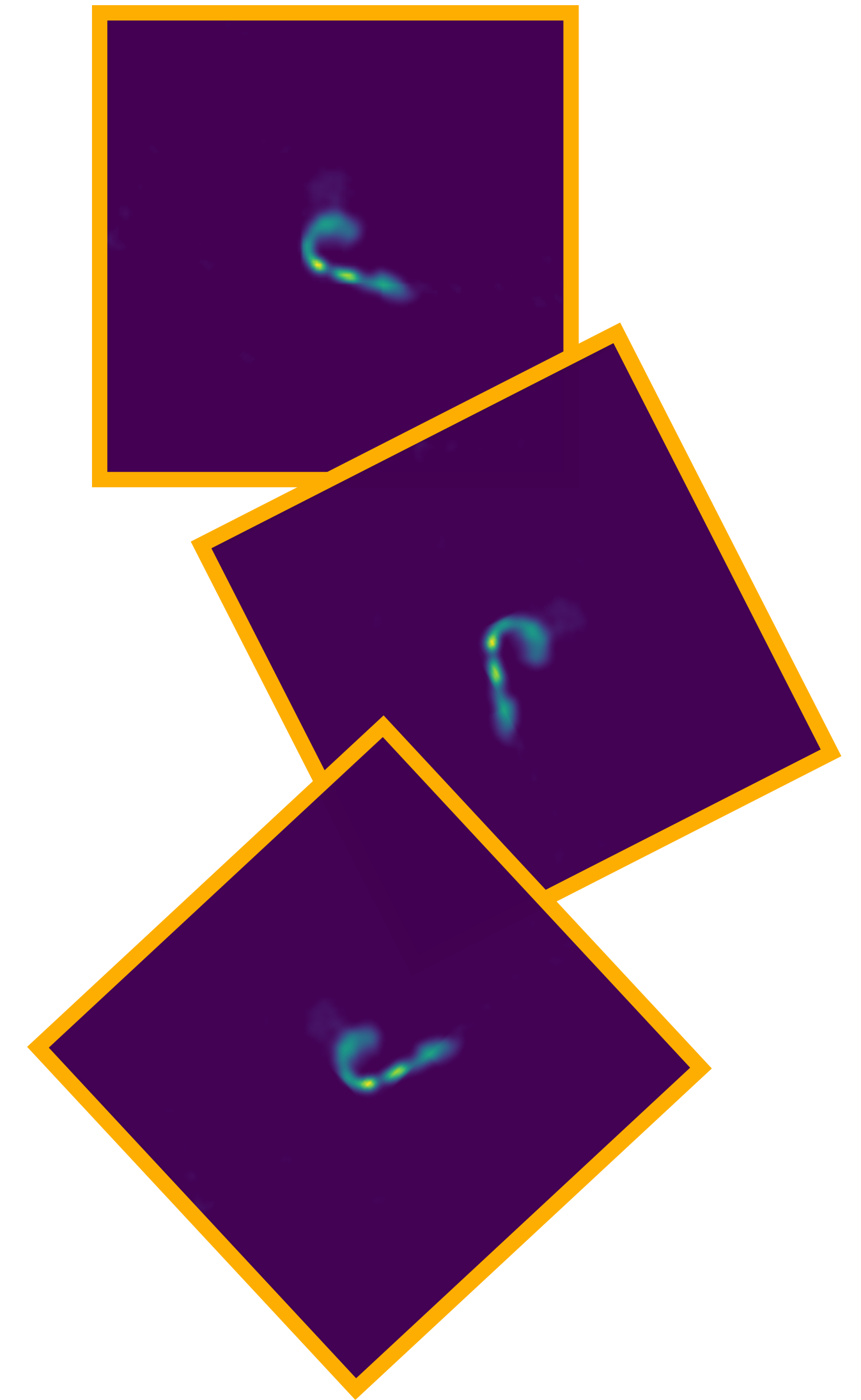
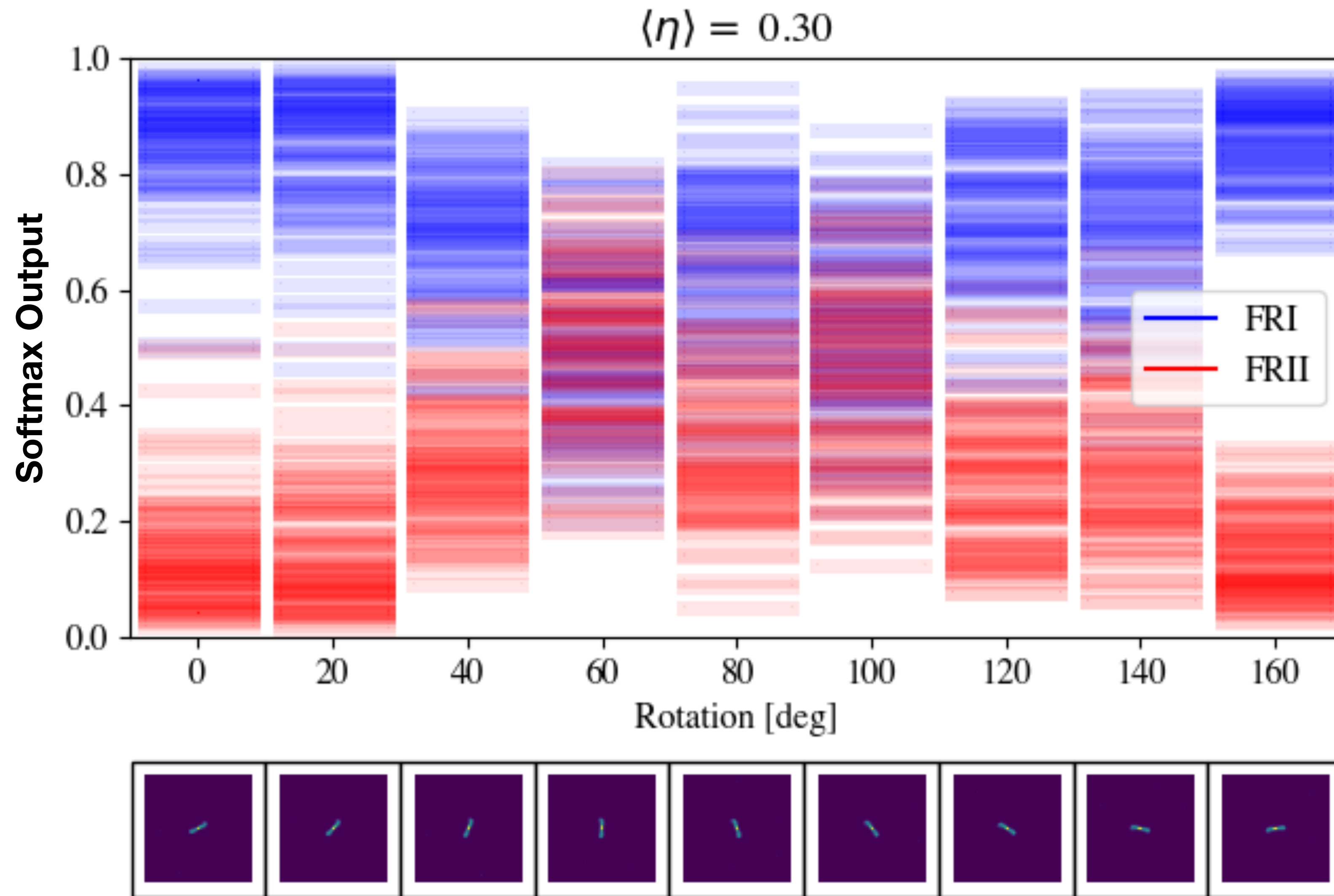
classifier

outputs



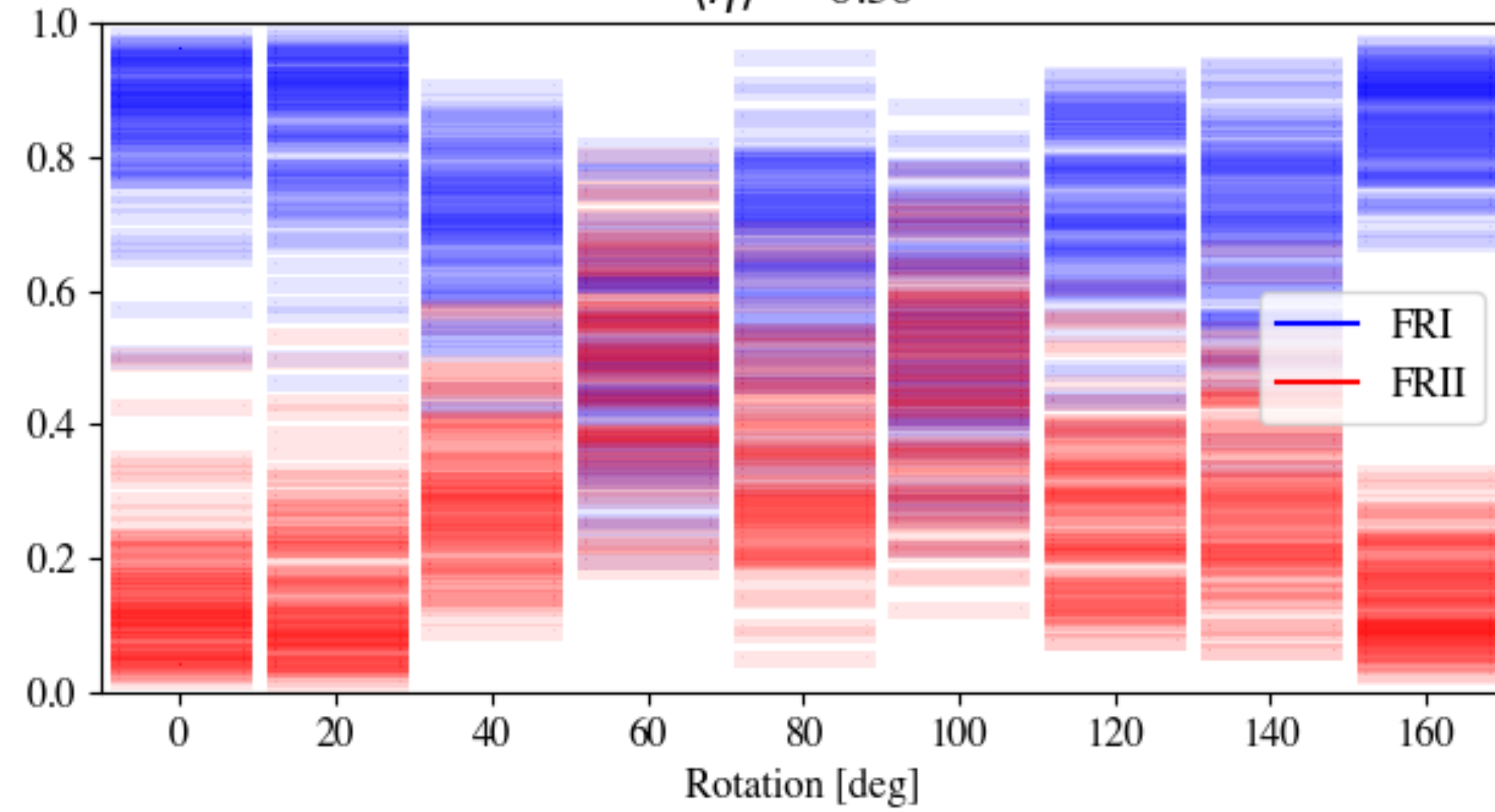
$$(\mathbb{R}^2, +)$$

$$(\mathbb{R}^2, +) \rtimes C_N$$



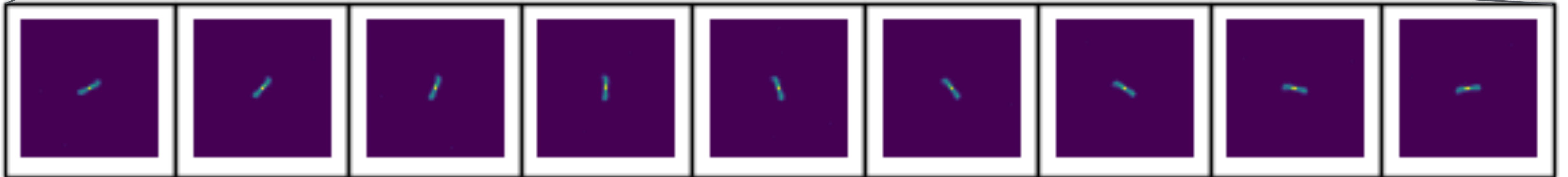
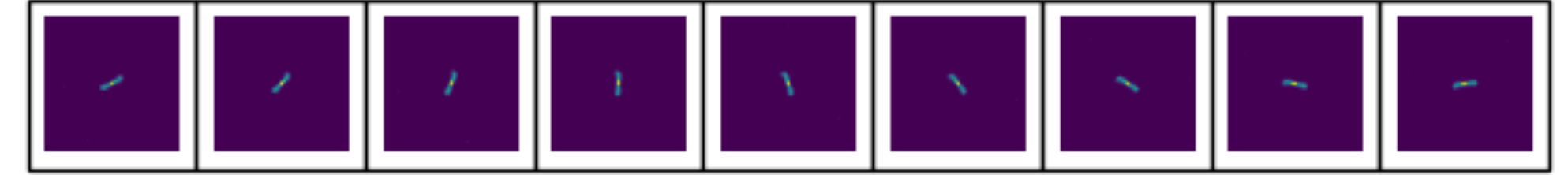
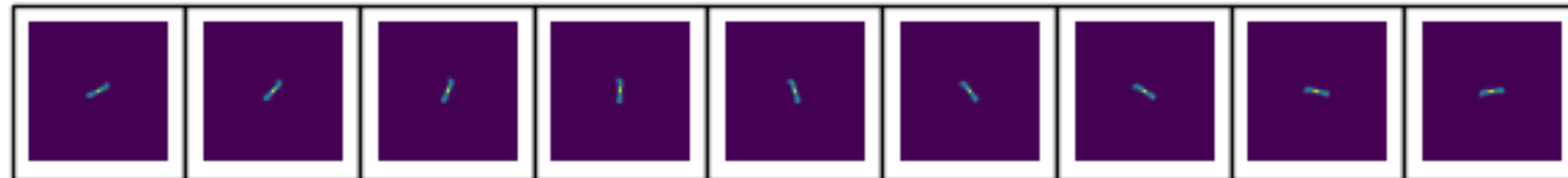
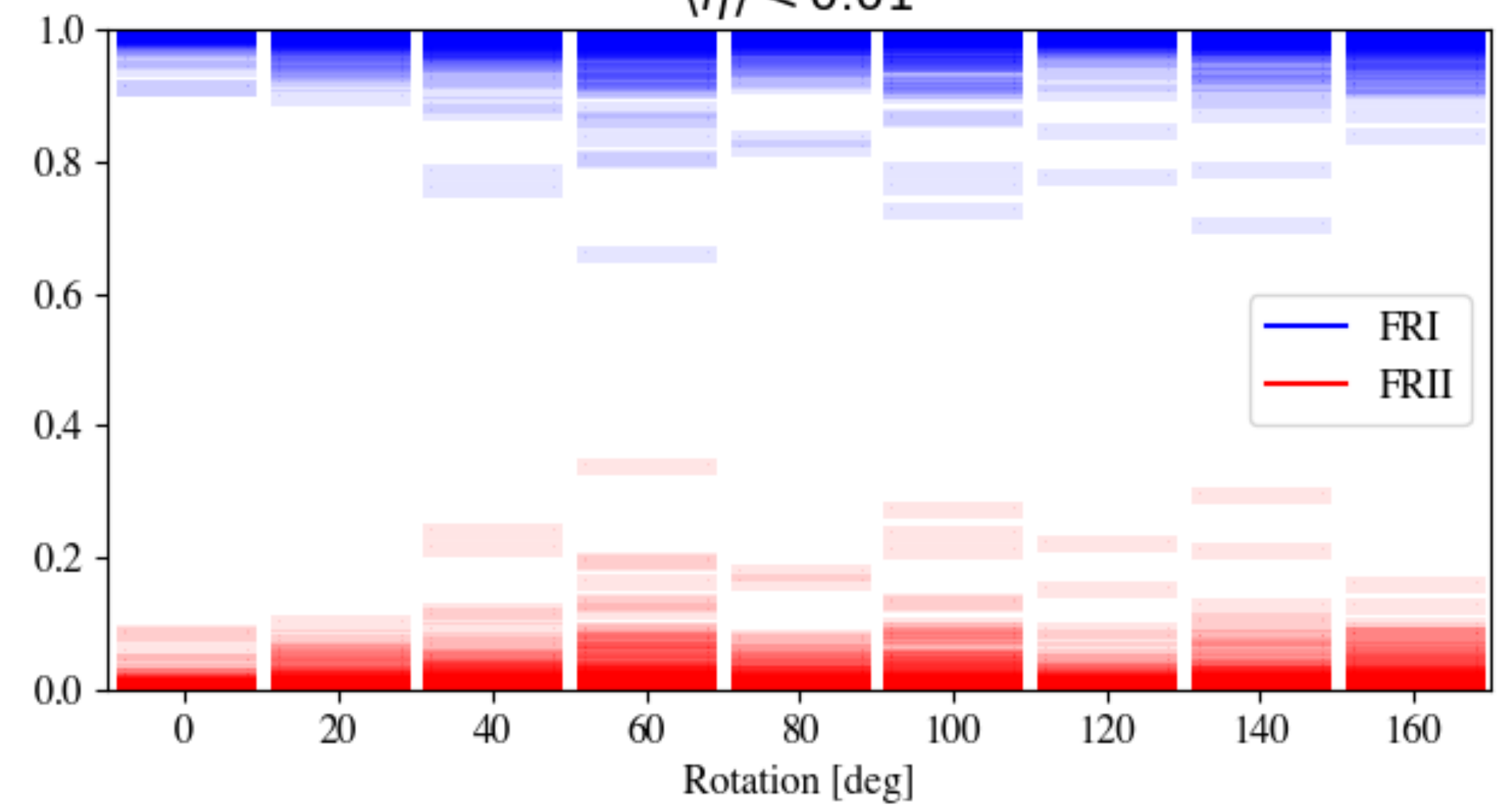
Conventional Convolutions, {e}

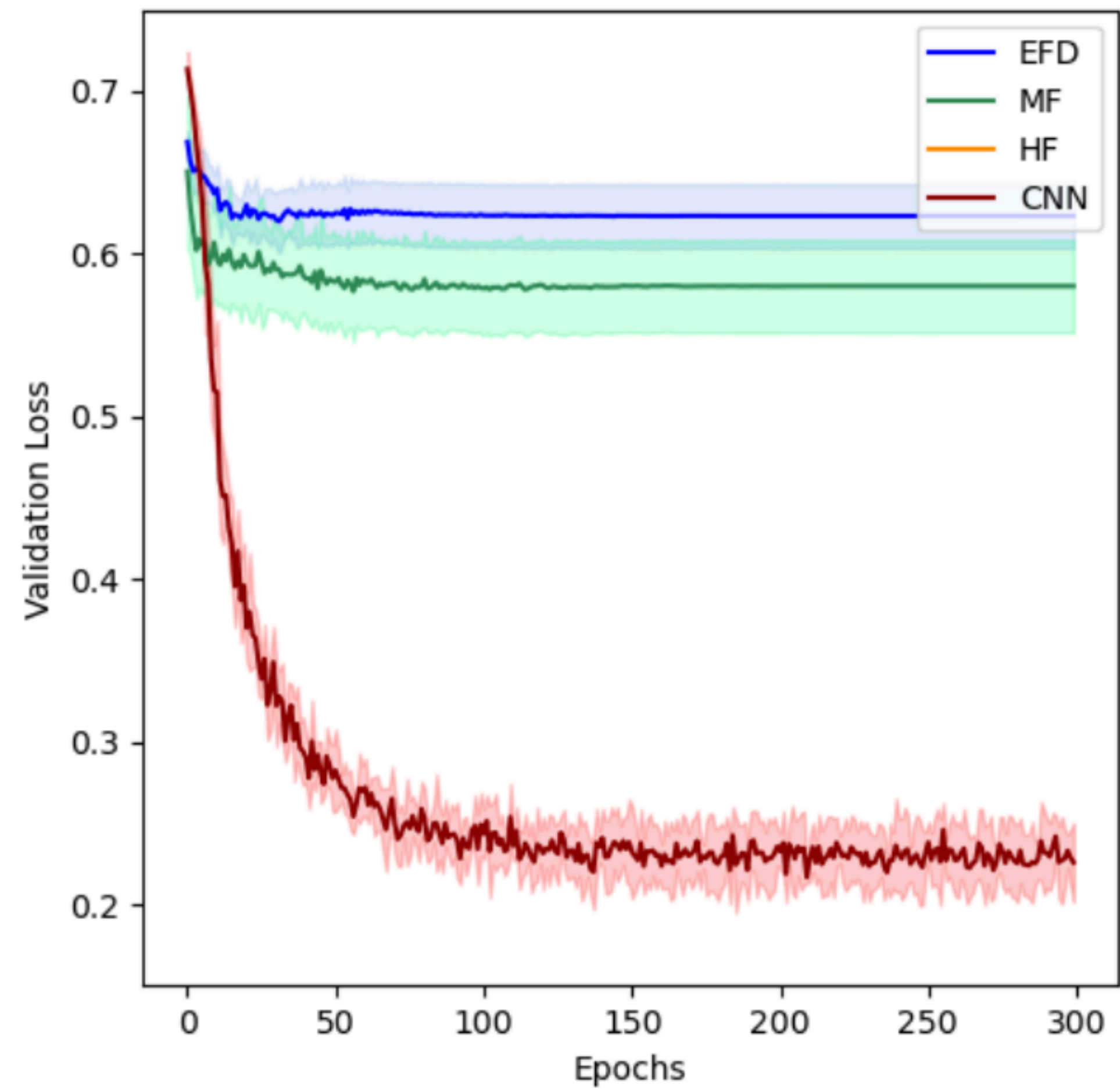
$$\langle \eta \rangle = 0.30$$



D_{16} Group-equivariant Convolutions

$$\langle \eta \rangle < 0.01$$





Features	Accuracy [%]	Computation Time [sec]
MF	67.98 ± 1.43	32.2 ± 0.6
HF	67.50 ± 1.54	43.5 ± 1.2
EFD	63.75 ± 3.04	85.9 ± 1.3
MF+HF	74.23 ± 1.28	21.8 ± 1.2
MF+EFD	67.50 ± 2.61	89.0 ± 1.5
HF+EFD	71.25 ± 2.41	64.3 ± 0.9
MF+HF+EFD	67.69 ± 1.37	39.7 ± 0.6
CNN	94.04 ± 1.37	2410.2 ± 7.7

Summary points

- Invariance and equivariance are important in astronomy
- When we use deep-learning approaches we should make sure that we are properly considering those requirements - and the potential for accidentally enforcing them inappropriately.
- There are multiple options available to implement group-equivariance explicitly into our deep-learning models.
- Empirically, explicit inclusion of group appropriate equivariance outperforms data driven vanilla approximations.

Equivariant and Coordinate Independent Convolutional Networks

What is the appropriate geometric structure for neural networks that process spatial signals on Euclidean spaces or more general manifolds? This question takes us on a journey which leads to a gauge field theory of convolutional networks.

Feature vector fields: The spatial signals we are interested in are fields of feature vectors. Feature fields allow to describe data like images, audio, videos, point clouds, or tensor fields, such as fluid flows and electromagnetic fields.

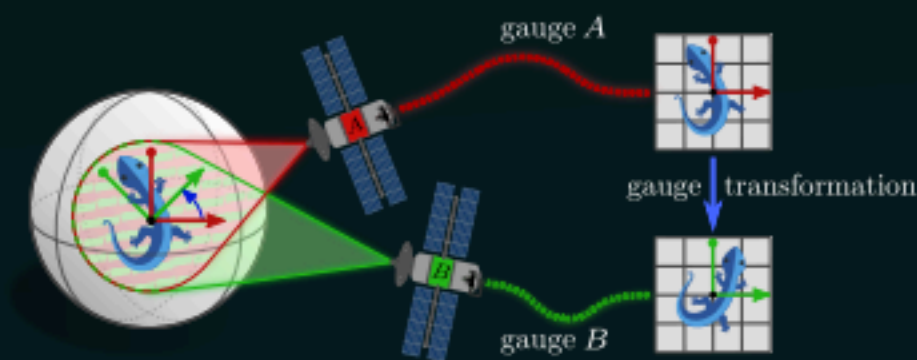
Equivariant networks commute with actions of some symmetry group on their feature spaces. The relevant group actions in this work are geometric transformations of feature fields, like translations, rotations, or reflections of images.

Equivariant models generalize everything they learn over the considered group of transformations. This property makes them significantly more data efficient, interpretable, and robust in comparison to non-equivariant models.

Convolutional Neural Networks (CNNs) are the most common network architecture for processing feature fields. Conventional CNNs operate on Euclidean spaces and are translation equivariant, i.e. position independent. This work explains how to extend CNNs to be equivariant under more general symmetries of space.

Coordinate independence: Manifolds are in general not equipped with a canonical choice of coordinates. Feature fields and neural network layers are hence required to be coordinate independent, that is, expressible relative to different frames of reference.

The ambiguity of local frames represents the gauge freedom of our neural field theory. We show that the demand for coordinate independence requires CNNs to be equivariant under local gauge transformations.



To offer an easy entry, the first part of this work focuses on the representation theory of equivariant convolutional networks on Euclidean spaces.

The insights gained in the Euclidean setting are subsequently leveraged to develop the full gauge theory of coordinate independent CNNs on Riemannian manifolds.

In the last part, we turn to a discussion of practical applications on specific manifolds. A comprehensive literature review demonstrates the generality of our theory by showing for more than 100 models from the literature how they can be understood as specific instantiations of "Equivariant and Coordinate Independent CNNs".

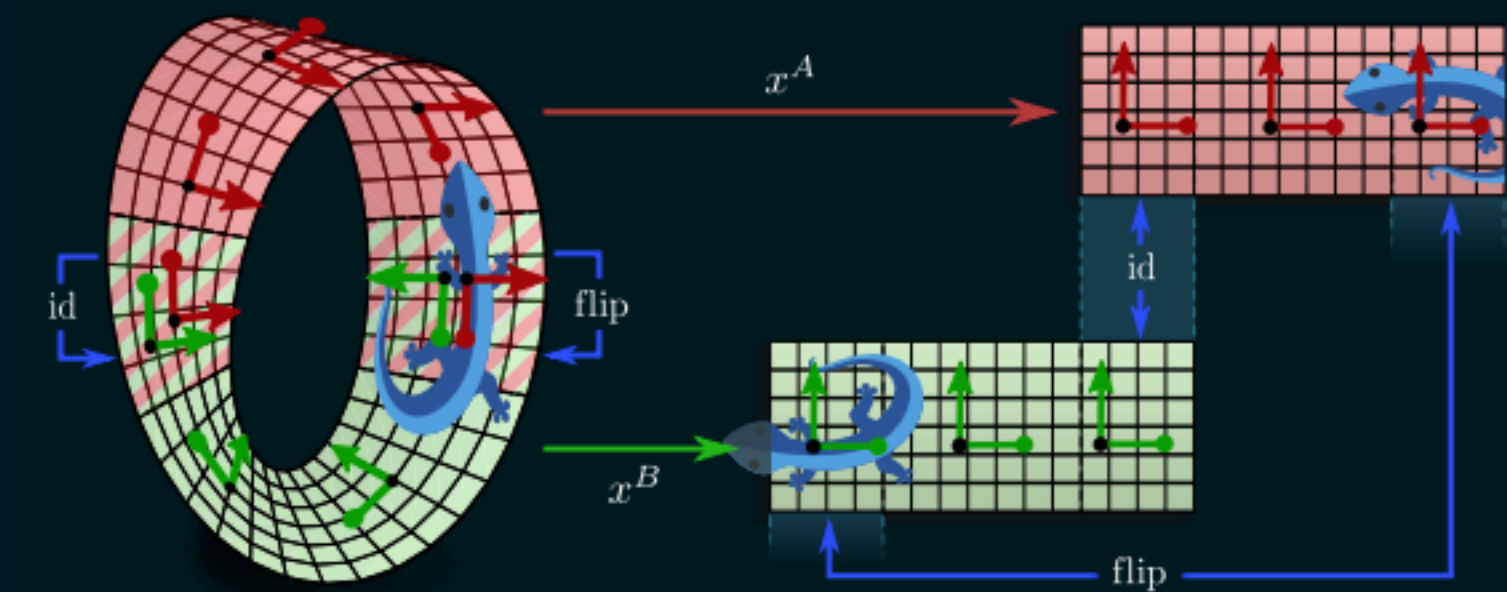
Equivariant and Coordinate Independent CNNs



Maurice Weiler

EQUIVARIANT AND COORDINATE INDEPENDENT CONVOLUTIONAL NETWORKS

A GAUGE FIELD THEORY OF NEURAL NETWORKS



Maurice Weiler
Patrick Forré
Erik Verlinde
Max Welling